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甘峰 的赞赏码

分析化学

Analytical Chemistry

甘峰



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Preface

Before retirement, I wrote two analytical chemistry textbooks: *Fundamentals of Analytical Chemistry* (ISBN: 978-7-5025-9676-7) and *Analytical Chemistry* (ISBN: 978-7-03-063519-8). When writing the former, I lacked experience; when writing the latter, I lacked time. The first was a case of not knowing how to write, and the second a case of not having time to write. Taken together, the two barely qualify as having been written.

Yet a traditional printed textbook is frozen the moment it goes to press. Even when the author discovers errors or hits upon a better way of explaining something, he can only wait for the next revision, and that cycle often stretches over years or even more than a decade. Writing a book is, in the end, an exercise in regret.

After retirement, my time is largely my own, and I have naturally lost my best excuse, that of being too busy. So an idea took hold: why not write, at a comfortable pace, a textbook that truly satisfies me? That is how this book came about.

This book is released in electronic form precisely to break free from the limitations of print. It can be revised, supplemented, and refined at any time, so the reader always receives the latest version. In this sense, it is not a finished book but a book that keeps growing; or rather, a book that can never be fully finished.

Writing a book is no easy task; it weighs on the mind and consumes no small amount of coffee. If you find this book helpful, especially if it spares you from failing a course, consider buying me a coffee ! Your support is what keeps me going.

This book is not without its shortcomings. For any errors or infelicities in the text, I would sincerely appreciate your corrections. My usual email addresses are:

cesgf@mail.sysu.edu.cn and ganfeng1964@163.com. I usually reply as promptly as I can; if I do not, I am most likely off somewhere pretending to survey historic sites, as the cover suggests.

GAN, Feng

Guangzhou, 2026 年 8 月 25 日

Contents

Cover	1
Preface	2
1 Introduction	16
1.1 What Is Analytical Chemistry?	17
1.2 The General Process of Analytical Chemistry	20
1.3 Classification and Selection of Analytical Methods	22
1.3.1 Technique, Method, Procedure, and Protocol	22
1.3.2 Classification by Method Principle	26
1.3.3 Classification by Analytical Task	28
1.3.4 Criteria for Choosing an Analytical Method	31
1.4 A Brief History of Analytical Chemistry	34
1.4.1 Analytical Chemistry in Antiquity	36
1.4.2 Analytical Chemistry in the Middle Ages	39
1.4.3 Analytical Chemistry in the Scientific Revolution	41
1.4.4 Analytical Chemistry in the Modern Era	45
1.4.5 Contemporary Analytical Chemistry	52
1.5 Application Fields of Analytical Chemistry	63
1.6 Is Analytical Chemistry a Science?*	69
Chapter Summary	74
Exercises	76

2 化学定量分析法概述	81
2.1 化学定量分析方法分类	81
2.1.1 重量分析法	82
2.1.2 滴定分析法	83
2.2 化学定量分析法与现代仪器分析的关系	84
2.2.1 重量法和滴定法是仪器分析的校准基石	84
2.2.2 化学定量分析法在当代分析任务中的不可替代性	85
2.2.3 从经典方法到现代方法的衔接	85
2.3 常用的量和单位	86
2.3.1 质量	87
2.3.2 体积	89
2.3.3 相对原子质量	90
2.3.4 相对分子质量	92
2.3.5 物质的量	93
2.3.6 摩尔质量	97
2.3.7 物质的量浓度	100
2.4 计量平衡关系	103
2.4.1 物质的量平衡	104
2.4.2 元素平衡	106
2.4.3 电荷平衡	108
2.4.4 电子数守恒	110
2.5 滴定分析法的基本概念	112
2.5.1 滴定过程及相关术语	113
2.5.2 标准溶液	113
2.5.3 对化学反应的要求与常用的滴定方式	114
2.5.4 滴定分析法的特点与分类	118

2.6	化学定量分析的一般步骤	119
2.6.1	采样	119
2.6.2	样品处理	121
2.6.3	样品的测量	123
2.6.4	结果的表示与评价	124
	本章要点总结	127
	习题	129
3	定量分析数据处理	135
3.1	误差从何而来	136
3.1.1	测量数据的获取	136
3.1.2	系统误差	138
3.1.3	随机误差	140
3.1.4	总误差	141
3.2	测量数据的统计规律	142
3.2.1	频数分布	142
3.2.2	正态分布	145
3.2.3	正态分布参数的含义	150
3.2.4	从样本推断总体	151
3.2.5	三大抽样分布简介	154
3.3	用统计工具处理实验数据	157
3.3.1	剔除离群值	158
3.3.2	表达测量结果	160
3.4	测量误差的传递规律	162
3.4.1	一般原理	162
3.4.2	加减法	164
3.4.3	乘除法	165

3.4.4	其它运算方法	166
3.5	有效数字及其运算规律	167
3.5.1	有效数字位数	168
3.5.2	有效数字的修约规则	169
3.5.3	有效数字的运算规则	170
3.5.4	有效数字运算中的一个约定 *	173
3.6	定量分析结果的评价	174
3.6.1	评价均值与真值	174
3.6.2	评价两组均值	179
3.6.3	假设检验中的两类错误	180
3.6.4	单因素方差分析 *	181
3.7	线性相关与回归	183
3.7.1	数学模型	183
3.7.2	一元线性回归	184
3.7.3	回归方程的显著性检验	186
3.7.4	预测区间	188
3.7.5	从 y 估计 x	190
	本章要点总结	194
	练习题	195
4	酸碱滴定法	199
4.1	酸碱质子理论与解离平衡	202
4.1.1	酸碱质子理论	202
4.1.2	酸解离常数	205
4.2	酸碱溶液中各种型体的分布	208
4.2.1	一元酸溶液中各种型体的分布	208
4.2.2	一元碱溶液中各种型体的分布	211

4.2.3	多元酸溶液中各种型体的分布	212
4.3	酸碱溶液中氢离子浓度的计算	216
4.3.1	一元强酸（碱）溶液中氢离子浓度计算	216
4.3.2	一元弱酸（碱）溶液中氢离子浓度计算	218
4.3.3	多元酸（碱）溶液中氢离子浓度计算	220
4.3.4	两性物质溶液中氢离子浓度计算	222
4.4	酸碱缓冲溶液	227
4.4.1	酸碱缓冲溶液的 pH 值计算	228
4.4.2	缓冲指数和缓冲范围	230
4.4.3	缓冲溶液的选择原则	231
4.5	酸碱滴定方程	232
4.5.1	强碱滴定一元强酸	233
4.5.2	强碱滴定一元弱酸	235
4.5.3	强碱滴定多元酸	239
4.5.4	酸碱滴定通式	241
4.6	滴定终点判定及滴定误差	246
4.6.1	酸碱指示剂的原理	247
4.6.2	理论变色点及终点误差	249
4.6.3	指示剂对滴定的影响 *	253
4.7	应用举例	256
4.7.1	凯氏定氮法测定有机物中氮含量	256
4.7.2	铵盐中含氮量的测定	262
4.7.3	烧碱中氢氧化钠和碳酸钠含量的测定	267
4.7.4	硅酸盐中二氧化硅含量的测定	269
4.7.5	皂化值的测定——返滴定法	273
4.8	酸碱滴定中二氧化碳的影响 *	277

本章要点总结	280
习题	281
5 配位滴定法	285
5.1 配位反应及相关参数	287
5.1.1 单齿配位反应	288
5.1.2 多齿配位反应	294
5.1.3 主反应与副反应	298
5.2 配位滴定方程	309
5.2.1 EDTA 滴定单一金属离子	309
5.2.2 EDTA 滴定多个金属离子	318
5.2.3 存在其它配体时滴定单一金属离子	324
5.3 混合离子的选择性滴定	325
5.3.1 近似判据	326
5.3.2 利用掩蔽剂提高滴定的选择性	333
5.3.3 采用其它配位滴定剂提高选择性	336
5.4 滴定终点判定及终点误差	337
5.4.1 金属离子指示剂的原理	338
5.4.2 理论变色点及终点误差	340
5.4.3 指示剂对滴定过程的影响 *	342
5.5 应用举例	346
5.5.1 水的总硬度	346
5.5.2 铜合金中铜锌含量测定	356
5.6 理论拓展 *	360
5.6.1 配位滴定产物浓度变化规律	361
5.6.2 1 : n 型配位滴定方程	365
5.7 本章要点总结	369

5.8 习题	371
习题	371
6 沉淀重量分析法	375
6.1 沉淀的类型及相关参数	378
6.1.1 沉淀的类型	378
6.1.2 沉淀的溶解度	379
6.1.3 活度、离子强度与溶度积的真实含义	381
6.1.4 条件溶度积	385
6.1.5 影响沉淀的溶解度的因素	388
6.2 沉淀的形成过程及影响纯度的因素	399
6.2.1 沉淀的形成过程	399
6.2.2 影响沉淀纯度的因素	404
6.2.3 沉淀形式、称量形式和表示形式	407
6.3 沉淀条件的控制	409
6.3.1 晶形沉淀的条件	410
6.3.2 无定形沉淀的条件	411
6.3.3 常用的沉淀剂	411
6.4 应用举例	413
6.4.1 废水中镁离子含量测定	413
6.4.2 氟化锂中硫酸根含量的测定	418
6.5 沉淀重量法的误差来源	423
本章要点总结	425
习题	427
7 沉淀滴定法	431
7.1 莫尔法	434

7.2	佛尔哈德法	440
7.2.1	直接滴定银离子	440
7.2.2	返滴定法测定卤素离子	442
7.3	法扬斯法	444
	三种银量法的比较	446
7.4	应用举例	448
7.4.1	天然水中氯含量的测定	448
7.4.2	银合金首饰中银含量的测定	450
7.5	本章要点总结	453
	习题	455
8	氧化还原滴定法	457
8.1	氧化还原反应及相关参数	458
8.1.1	氧化还原反应举例	459
8.1.2	能斯特方程与表观电极电位	462
8.1.3	氧化还原反应平衡常数	464
8.1.4	影响氧化还原反应进程的因素	467
8.2	氧化还原滴定方程	469
8.2.1	以浓度为指标	470
8.2.2	以电位为指标	474
8.2.3	化学计量点与滴定突跃范围	477
8.3	滴定终点的判定与滴定误差	479
8.3.1	氧化还原指示剂的作用原理	479
8.3.2	终点误差	481
8.4	待测体系的预氧化或预还原	482
8.4.1	常用的预氧化剂	482
8.4.2	常用的预还原剂	484

8.5	几种典型的氧化还原滴定法	486
8.5.1	高锰酸钾法	486
8.5.2	铈量法	490
8.5.3	重铬酸钾法	490
8.5.4	碘量法	494
8.5.5	溴酸钾法	500
8.6	应用举例	501
8.6.1	铁矿石中全铁含量的测定	501
8.6.2	工业废水中化学需氧量的测定	503
8.6.3	铜精矿中铜含量的测定	506
8.7	本章要点总结	509
	习题	511
9	吸光光度法	516
9.1	电磁辐射与吸收光谱的基本原理	517
9.1.1	电磁辐射的波粒二象性	518
9.1.2	分子能级与光谱类型	519
9.1.3	分子中的电子跃迁类型	520
9.1.4	生色团与助色团	522
9.1.5	无机化合物的吸收 *	524
9.1.6	吸收光谱与颜色	526
9.1.7	紫外-可见吸收光谱的定性分析应用 *	527
9.2	朗伯-比尔定律	531
9.2.1	透光度与吸光度	531
9.2.2	朗伯-比尔定律的推导	532
9.2.3	吸光度的加和性	535
9.3	朗伯-比尔定律的偏离	536

9.3.1 化学性偏离 536

9.3.2 仪器性偏离 538

9.3.3 吸光度测量的误差 542

9.4 分光光度计 544

9.4.1 分光光度计的基本结构 544

9.4.2 检测器噪声与浓度测量误差 555

9.4.3 积分球检测器与固体样品测定附件 561

9.4.4 分光光度计的光路类型 564

9.4.5 分光光度计的主要检定项目 567

9.5 分析条件的选择 570

9.5.1 溶液体系的分析条件 570

9.5.2 固体样品的分析条件 573

9.6 吸光光度法的定量分析技术 575

9.6.1 单组分的定量分析 576

9.6.2 标准加入法 577

9.6.3 多组分的同时定量分析 578

9.6.4 导数光谱法 581

9.6.5 分光光度滴定 582

9.6.6 示差光度法 583

9.7 应用举例 587

9.7.1 邻二氮菲分光光度法测定水中铁 588

9.7.2 二苯碳酰二肼分光光度法测定水中六价铬 590

9.7.3 纳氏试剂分光光度法测定氨氮 592

9.8 应用拓展 * 594

9.8.1 流动注射-光度联用技术 594

9.8.2 配合物组成与稳定常数的测定 595

9.8.3 固体样品的透射与反射测定 599

9.8.4 方法评价 602

本章要点总结 603

习题 603

10 分离与富集方法 608

10.1 概述 610

10.2 沉淀与共沉淀分离法 612

10.2.1 沉淀分离法 612

10.2.2 共沉淀分离法 616

10.3 蒸馏与挥发分离法 619

10.4 萃取分离技术 621

10.4.1 萃取分离技术概述 621

10.4.2 超临界萃取 628

10.4.3 固相萃取 629

10.4.4 固相微萃取 630

10.4.5 液相微萃取 634

10.5 离子交换分离法 634

10.5.1 离子交换剂的种类 634

10.5.2 离子交换剂的结构 635

10.5.3 离子交换剂的性质 637

10.5.4 离子交换平衡 639

10.5.5 离子交换分离操作方法 640

10.5.6 离子交换分离法的应用 641

10.6 色谱分离法 642

10.6.1 分离的通用理论 643

10.6.2 柱色谱分离法 647

Contents	15
10.6.3 纸色谱分离法	648
10.6.4 薄层色谱法	650
本章要点总结	652
本章要点总结	652
习题	654
附录 1 基本常数	657
附录 2 程序示例	679
B1 Octave 安装	679
B2 定量分析数据处理	685
B2.1 剔除离群值	685
B2.2 一元线性回归	689
B3 酸碱滴定有关的程序	693
B4 配位滴定有关的程序	701
Answer	713
Index	715
References	717

Chapter 1 Introduction

Learning Objectives

- Understand the definition of analytical chemistry and distinguish the concepts of “analytical chemistry” and “chemical analysis”.
- Master the general workflow of analytical chemistry: sampling → sample preparation → measurement → data processing → reporting of results.
- Understand the distinctions among technique, method, procedure, and protocol.
- Master the classification of analytical chemistry by method principle, by analytical task, and by field of application.
- Understand the key criteria for choosing an analytical method (accuracy, sensitivity, selectivity, cost, etc.).
- Understand the historical development of analytical chemistry, from antiquity to the present, and become familiar with its pivotal figures and events.
- Recognize the central roles of analytical chemistry in drug development, food safety, environmental monitoring, and clinical diagnosis.

In March 2008, the First Hospital of the Chinese People’s Liberation Army began admitting a series of infants suffering from “kidney stones,” and similar cases soon appeared in Gansu, Jiangsu, Ningxia, and other regions. Investigation revealed that the culprit was melamine, which unscrupulous merchants had illegally added to milk powder. Under the national standard in force at the time, the protein content of milk powder was determined by the classic Kjeldahl method, which infers protein content by converting nitrogen-containing substances in

the sample into ammonium salts and measuring the nitrogen content. Because melamine contains about 66% nitrogen and is likewise converted to ammonium salts by the Kjeldahl method, lawbreakers were able to conceal the adulteration and fabricate a false impression of high protein content. Analytical chemists acted quickly, using high-performance liquid chromatography (HPLC) to confirm the presence of melamine in the milk powder of numerous manufacturers. In the end, this incident left more than 290,000 infants with abnormalities of the urinary system, and six infants died. It makes us realize with clarity that analytical chemistry is by no means an ivory-tower discipline confined to the laboratory; it is a subject bound up with the life and health of every individual and with the public safety of society. This is precisely what makes the study of analytical chemistry meaningful.

1.1 What Is Analytical Chemistry?

Before beginning the formal study of analytical chemistry, it is essential to form an accurate picture of the discipline itself, for this will help the reader to approach the theory with the right mindset. Most people encountering this subject for the first time have probably seen, in films, on television, or in other media, laboratory personnel applying one analytical measurement technique or another to some sample, and so they unconsciously assume that analytical chemistry is nothing more than the technical work those laboratory workers perform. In fact, what people commonly see and call analytical chemistry should more strictly be called chemical analysis. Chemical analysis refers to the use of existing analytical methods to solve qualitative and quantitative analysis problems encountered in routine work.

Then what is analytical chemistry? Here we offer a definition in Chinese and

a definition in English, as follows:

Definition 1.1

Analytical chemistry is a science that develops and applies various methods, instruments, and strategies to obtain information about the composition and properties of matter, within the dimensions of time and space.

Definition 1.2

Analytical chemistry is the science that investigates the development and application of methods, instruments and strategies to find the composition of materials.

The Chinese definition is the one commonly adopted in China, while the English definition derives from the classic exposition of the discipline by the International Union of Pure and Applied Chemistry (IUPAC). The definitions above may appear concise, yet they cover the core task of analytical chemistry, namely obtaining information about the composition and properties of matter, which is embodied concretely in qualitative analysis, quantitative analysis, structural analysis, and the like. At the same time, both definitions emphasize the dynamic and innovative character of analytical chemistry: it is not a fixed operating manual, but a discipline that continuously absorbs new principles and new techniques in order to extend the boundaries of its capabilities. To develop a deeper understanding, we now examine the essential elements of the definitions above one by one.

First, the definitions explicitly characterize analytical chemistry as a science. This characterization actually distinguishes “analytical chemistry” from “chemical analysis”: the latter readily calls to mind the use of various instruments and the dazzling display of spectral and chromatographic results, whereas the former

centers on the abstract theoretical system underlying the discipline, which is precisely the intrinsic attribute that allows analytical chemistry to stand as an independent branch of science. Making this point explicit reminds the reader that one of the aims of this book is to expound systematically the theoretical dimension of analytical chemistry. To be sure, the book also treats experimental content (based mainly on national standard methods), but more often it explains and discusses matters from a theoretical point of view.

Second, the definitions above position analytical chemistry as a discipline in a continuous process of evolution, which is entirely consistent with its intrinsic nature. In fact, analytical chemistry has always been a discipline at the frontier. Time and again humanity has discovered that, when venturing into new fields of nature or new directions within existing fields, one inevitably encounters new analytical chemistry problems, and new analytical methods must therefore be constructed. Even within everyday analytical chemistry, as demands on accuracy and sensitivity rise, established methods must be improved, and entirely new methods may need to be developed.

Finally, the definitions above confer upon analytical chemistry a mission, namely obtaining information about matter. Within the traditional framework, obtaining information about matter manifests itself in determining the chemical composition of a substance, measuring the content of each constituent, and characterizing chemical structures and forms. Analytical chemistry has accordingly given rise to numerous branches, including qualitative analysis, quantitative analysis, structural analysis, and speciation analysis. Under the modern framework, however, far higher demands are placed on “obtaining information about matter.” In medicine, for example, the discovery of biomarkers that trigger a disease is crucial to its treatment, so establishing analytical methods capable of discover-

ing such biomarkers is of particular importance.

1.2 The General Process of Analytical Chemistry

A complete analytical undertaking follows a logical chain from question to answer, and can generally be divided into six major stages (Figure 1.1).

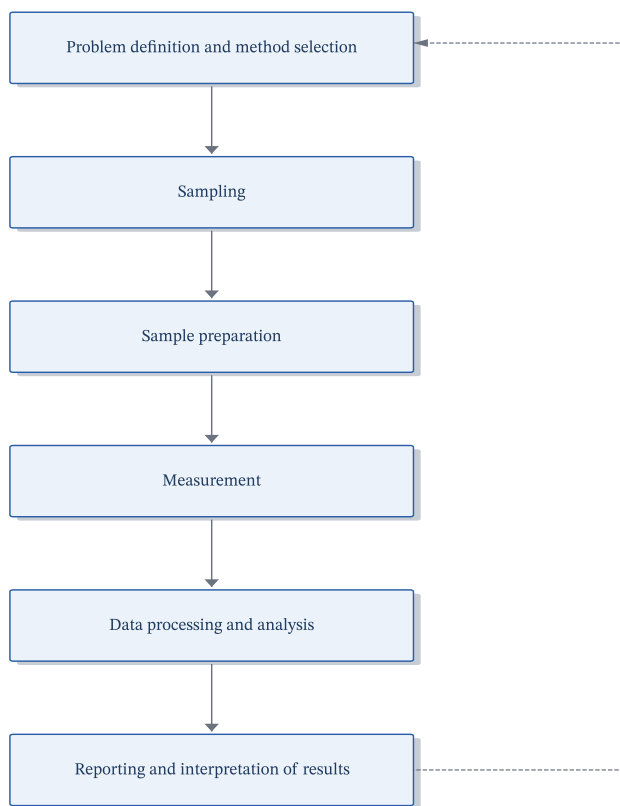


Figure 1.1: The general process of analytical chemistry

Stage 1: Problem definition and method selection. Before any experimental operation, one must first clarify “what is to be analyzed?” “why is it to be analyzed?” and “for what purpose will the result be used?” The answers to these questions determine the direction of all subsequent stages. For example, does the determination of lead in drinking water require sensitivity at the parts-per-billion

(ppb) level? This requirement directly determines which analytical methods are viable. The selection of analytical methods is discussed in detail in Section 1.3.4.

Stage 2: Sampling. Sampling is the most easily overlooked, and possibly the most error-prone, stage of the entire analytical process. Whether one samples from an ore body in a mine, the water of a lake, or the body of a patient, the sole core principle is to ensure that the sample taken truly reflects the overall state of the parent population. Specific sampling methods and their theory are discussed in detail in Chapter 2.

Stage 3: Sample preparation. The raw sample collected is often not directly measurable; it must undergo digestion, removal of interferences, enrichment, and other preparatory steps. These operations are collectively called sample preparation, and the specific methods are introduced in Chapter 2.

Stage 4: Measurement. This is the core stage of the analytical undertaking. Some analytical technique is applied directly to measure a physical or chemical property of the analyte, yielding the raw signal. The specific requirements of measurement and error control in chemical quantitative analysis are discussed in Chapter 2.

Stage 5: Data processing and analysis. Raw measurement signals (mass, volume, absorbance, potential difference, etc.) must be converted into the concentration or content of the analyte. For gravimetric and titrimetric methods, mass and volume signals can be converted directly into analyte content through stoichiometric relationships; for instrumental methods, signals such as absorbance and potential difference must be converted into concentration or content by means of calibration curves, standard addition, and similar approaches. At the same time, the reliability of the results must be assessed, that is, error analysis and uncertainty evaluation must be performed.

Stage 6: Reporting and interpretation of results. The final analytical result must be checked against the original problem, answering “how much of the analyte is present in the sample?” “does the result meet the required accuracy?” and “what conclusions does it support?” If the result is not as expected, one must backtrack (feedback and iteration) to trace the source of the problem and optimize the method.

In actual work, the six stages above are not completed in a single one-way pass. The analyst may discover anomalies in data processing after measurement and return to modify the sample preparation procedure; or may find during reporting that the problem was not clearly defined and replan the entire analytical strategy. This feedback and iteration mechanism is precisely the salient feature of analytical chemistry as a highly practical discipline.

1.3 Classification and Selection of Analytical Methods

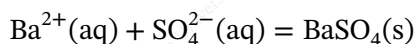
Before discussing the classification and selection of analytical methods, we must first clarify four basic concepts that are progressively nested and run through the entire book: technique, method, procedure, and protocol.

1.3.1 Technique, Method, Procedure, and Protocol

In analytical chemistry, **technique** (technique), **method** (method), **procedure** (procedure), and **protocol** (protocol) are four progressively nested concepts, and understanding their distinctions precisely helps to build a systematic way of analytical thinking.

Measurement in analytical chemistry is based primarily on chemical and physical principles. These principles constitute the **technique** for analyzing a sample. For example, barium ions react with sulfate ions to form a barium sul-

fate precipitate:



This chemical reaction principle constitutes the possibility of determining barium ions (or sulfate ions) in solution. But with this technique alone, one still cannot analyze a concrete sample directly.

To apply a technique to a concrete chemical analysis, an analytical **method** must be constructed for the specific sample. An analytical method is a set of analytical steps, built for the particular matrix of a certain type of sample, that is dedicated to determining a given analyte in that sample. When the same principle is applied to different samples, the methods adopted are usually not identical. For example, although the barium sulfate precipitation principle can be used to determine barium content, the specific methods will differ depending on whether the barium ions are in soil, water, or body fluids. When we discover a new chemical or physical principle, it constitutes only a technique, and cannot yet be applied directly to the analysis of a concrete sample. Transforming a chemical or physical principle into a practically usable analytical method requires extensive and deep research by scientists.

An established analytical method usually enjoys relatively broad applicability. Different institutions using such methods will, according to their own specific needs, develop detailed guiding principles, and such a guiding principle is called a **procedure**. Besides specifying in detail how to apply the method to a particular analyte, a procedure must also specify in detail how to sample, how to deal with interferences, and how to evaluate the measurement results.

A procedure formally compiled by an official body of some kind is called a **protocol**. Protocols are usually established by administrative agencies, and their content consists of strict written directives specifying procedures that must be

strictly followed. For example, a laboratory must have a protocol for laboratory safety, a protocol for using analytical instruments, and so on. Each protocol must focus on a specific aspect. When multiple laboratories jointly carry out a survey, all of them must follow the same protocol.

Key Concept

In short, a technique is an analytical principle, a method is the concrete implementation of a technique, a procedure is a guide to applying a method in a particular setting, and a protocol is a procedure that has been standardized and officially recognized. The four constitute a complete chain from theory to practice.

Self-Check

True or false:

1. A technique can be applied directly to the analysis of any sample containing the analyte of interest. ()
2. When the same analytical method is used by different laboratories, their procedures may differ. ()
3. A protocol is an internal document drawn up by an individual laboratory, and is not binding. ()

Hint: Review the core definitions and the differences in binding force of the four levels in Table 1.1.

Having understood the hierarchical relationship among technique, method, procedure, and protocol, we can now examine the classification of analytical chemistry. The classifications are numerous and become increasingly refined as research and application fields expand. How are the existing analytical methods classified, and for which scenarios is each type suitable?

Table 1.1: A comparison of technique, method, procedure, and protocol

Concept	English	Core definition	Example
Technique	technique	A chemical or physical analytical principle that provides the possibility of analysis	Ba^{2+} reacting with SO_4^{2-} to form a barium sulfate precipitate
Method	method	A concrete analytical scheme constructed for a specific matrix and analyte	Gravimetric determination of barium in soil
Procedure	procedure	An internal operating guide compiled by an institution according to its own equipment and personnel conditions	A soil barium determination work instruction issued by an environmental monitoring station
Protocol	protocol	A standardized text promulgated by an authoritative body, partly with statutory or industry-binding force	GB (mandatory national standard), GB/T (recommended national standard), ISO international standards, EPA methods

1.3.2 Classification by Method Principle

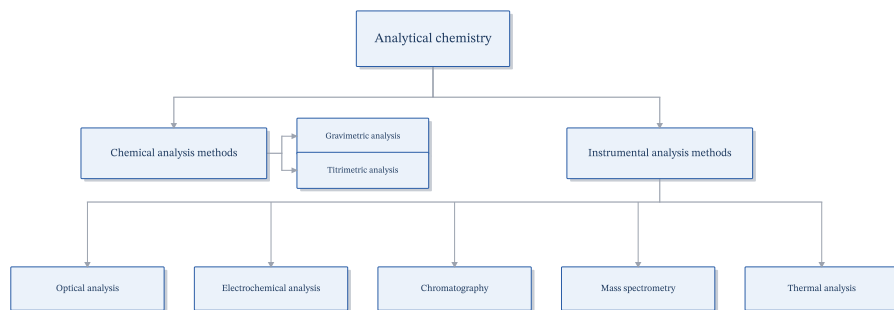


Figure 1.2: Classification of analytical chemistry by method principle

According to method principle, analytical chemistry is divided primarily into chemical analysis methods and instrumental analysis methods, as shown in Figure 1.2. Chemical analysis methods are based on well-established chemical reactions, and their quantitative basis is the strict stoichiometric relationship between reactants and products. Early analytical methods mostly belonged to this class, so chemical analysis forms the core of classical analytical chemistry. Its outstanding advantage is that only two basic physical quantities need to be measured precisely, namely mass and volume, and both have direct and authoritative international standards. Under certain conditions, chemical analysis is therefore also called an absolute method. An absolute method is a quantitative method that can produce results using only fundamental physical constants and stoichiometric relationships, without calibration by reference materials. The detailed classification and theoretical foundations of chemical quantitative analysis will be expounded systematically in Chapter 2.

Instrumental analysis methods are based on the physical and physicochemical properties of matter, and are so named because they require special instruments. According to their physical principles, instrumental analysis methods are

further divided into optical analysis, electrochemical analysis, thermal analysis, mass spectrometry, chromatography, and so on. Instrumental analysis methods are characterized by high sensitivity, good selectivity, and rapid analysis, and are the methods of choice for determining trace and ultratrace constituents.

Germany offers an instructive example. In the 1980s, with the rapid development of agriculture and the increasingly widespread use of pesticides, food safety problems gradually drew public concern. Against this background, fluorescence spectroscopy, an optical technique, attracted the interest of researchers, who eventually established a method for detecting pesticide residues in vegetables such as lettuce and broccoli. The vegetable samples were washed and processed into standard test specimens, then placed in a specially designed detection chamber. When the specimen was irradiated with ultraviolet light of wavelength 365 nm, specific chemical bonds in the organophosphorus pesticide molecules remaining in the vegetables absorbed the light energy and emitted characteristic fluorescence (the emission peaks of different organophosphorus pesticides generally fall in the range 380 ~ 500 nm). Even at residue levels as low as a few micrograms per kilogram, the technique still achieved effective detection. The method was subsequently adopted by German food regulatory authorities, which accordingly formulated stricter national standards for pesticide residues in vegetables and promoted their application throughout the country.

In modern analytical chemistry, instrumental analysis has become the mainstream research tool, and new methods and new instruments are continually emerging, greatly accelerating the development of the discipline. By comparison, many traditional chemical analysis methods are gradually being replaced by instrumental analysis. Some have even proposed replacing the term “chemical analysis” with “physical analysis,” which poses a new challenge to the value of

analytical chemistry as an independent discipline. Yet instrumental analysis still has limitations and cannot completely replace chemical analysis. For example, instrumental analysis usually requires reference standards to calibrate the measurement signal. If no reference standard is available, the analysis can hardly proceed; and if the reference standard is expensive, the cost of analysis rises significantly. Moreover, in certain applications, such as samples of relatively simple composition and relatively high analyte content, chemical analysis can yield accurate and reliable results with simpler and more economical operations. Instrumental analysis and chemical analysis are therefore not mutually exclusive but complementary; in practice, the method should be chosen flexibly according to the specific need, and neither should be neglected.

1.3.3 Classification by Analytical Task

According to the analytical task, traditional analytical chemistry is divided into qualitative analysis, quantitative analysis, and structural analysis. The task of qualitative analysis is to discover which elements, atomic groups, or compounds are present in a sample. Qualitative analysis is often the first step in chemical analysis; only after it is known which components are present in a sample can subsequent analytical work proceed. In archaeology, for example, when an ancient bronze artifact is unearthed, conservators first apply qualitative analysis. By scanning the surface of the bronze with X-ray fluorescence spectrometry, the instrument excites characteristic X-rays of the elements contained in the bronze, thereby determining the major elements such as copper, tin, and lead, as well as trace impurity elements such as arsenic and nickel. This provides key basic information for studying the manufacturing techniques, dating, and conservation schemes of the artifact.

Quantitative analysis is the determination of the content of an analyte of interest in a sample. Relative to qualitative analysis, quantitative analysis is more complex and more demanding. For example, in clinical treatment it is sometimes necessary to know accurately the concentration of potassium in a patient's blood, and to decide on this basis whether to administer potassium or to take measures to lower it; otherwise the patient's life may be endangered. As another example, in steel production the carbon content of the molten steel in the furnace must be controlled precisely. Using an infrared carbon-sulfur analyzer, the steelmaker rapidly cools a sample of the molten steel into a thin disk, places it in the instrument, and, exploiting the absorption of specific infrared wavelengths by carbon, determines the carbon content quickly and accurately. For instance, if the measured carbon content is 0.35%, one must then judge whether coke or other raw materials should be added to adjust the composition of the molten steel, so as to ensure that the steel meets the strength, toughness, and other performance requirements.

The task of structural analysis is to determine the molecular structure of the analyte, providing spatial structural information for understanding chemical reactions at the molecular level, or structural information for the biological activity of molecules. In drug development, for example, researchers use nuclear magnetic resonance (NMR), X-ray single-crystal diffraction, and other techniques for structural analysis of newly synthesized compounds in the search for anticancer drugs. When studying a potential small-molecule targeted anticancer drug, NMR provides information on the chemical environments of the hydrogen and carbon atoms in the molecule, and X-ray single-crystal diffraction yields the precise three-dimensional structure; together they identify the key active site at which the drug molecule binds to the target protein of the cancer cell, providing a solid basis for

subsequent optimization of the drug structure and improvement of its efficacy, and helping the candidate drug move from the laboratory toward clinical application.

In recent years, driven by the needs of scientific research and production, new categories such as speciation analysis and energy-state analysis have been proposed. Speciation analysis, for example, arose from the growing recognition that different chemical forms of an element exhibit distinct properties, with consequences for environmental chemistry, nutrition, pharmacology, and the work of chemists and biologists; it focuses on how the chemical form of an element affects its biological activity or environmental behavior.

For instance, mercury ions are highly toxic to organisms, and the toxicity of methylmercury is far greater still than that of inorganic mercury. Moreover, methylmercury is highly lipophilic and biocompatible, readily binds to biological tissues, and is thus transmitted and enriched step by step along the food chain. The “Minamata disease” incident of the 1950s in Minamata, Kumamoto Prefecture, Japan, originated from mercury in industrial discharges that was converted to methylmercury by microorganisms in the sediment; the methylmercury was taken up and accumulated by fish, and local residents who ate the contaminated fish over a long period were poisoned, bringing profound disaster to the health of the population.

As another example, speciation analysis of arsenic in soil is crucial in environmental monitoring. Researchers use high-performance liquid chromatography coupled with inductively coupled plasma mass spectrometry (HPLC-ICP-MS) to analyze soil samples collected near mining areas after appropriate sample preparation. The technique separates different arsenic species by their differing retention times on the chromatographic column, and the mass spectrometer then

determines precisely the content of each arsenic species. Studies have found that certain organic arsenic species, such as dimethylarsinic acid, although accounting for only a small fraction of total arsenic, are more mobile and may enter plants more readily than inorganic arsenic species under certain conditions, and may enter the human body through the food chain, posing a potential health risk. This finding has promoted the development of targeted soil remediation techniques, such as planting specific plants that absorb organic arsenic to reduce the risk of soil arsenic contamination.

In the area of energy-state analysis, researchers working on battery development for new-energy vehicles perform energy-state analysis of electrode materials for lithium-ion batteries. Using photoelectron spectroscopy to probe the distribution of electronic states on the electrode material surface, they found that the surface density of electronic states of a cathode material treated with a special coating changed, and the electron migration rate increased markedly. At the same time, electrochemical impedance spectroscopy was used to monitor the energy-state changes during charge-discharge, revealing that in the optimized battery system the energy barrier for lithium-ion intercalation into the electrode material is lowered, greatly improving fast-charging performance: the full charge time was reduced from 2 hours to 40 minutes. This provides powerful technical support for the development of new-energy vehicles.

1.3.4 Criteria for Choosing an Analytical Method

Faced with a concrete analytical problem, how does one choose the most suitable method from the many available? This is a core skill that every analytical chemist must master. The choice of an analytical method usually requires a comprehensive trade-off among the following dimensions:

Accuracy and precision. Accuracy reflects how close the measured value is to the true value, and precision reflects how scattered the results of repeated parallel measurements are. For forensic evidence identification or the determination of drug components, high accuracy is the primary requirement; whereas for trend analysis in environmental monitoring, high precision may matter more than absolute accuracy.

Sensitivity and detection limit. Sensitivity measures the rate at which the instrument response changes with the concentration of the analyte, while the detection limit defines the lowest concentration the method can reliably detect. In the analysis of ultrapure materials and of trace environmental pollutants, a low detection limit is often the decisive factor in method selection.

Selectivity. A highly selective analytical method can identify and respond to the target analyte alone in a complex matrix, without significant interference from coexisting substances. In practice, selectivity is often the more difficult requirement to meet than sensitivity.

Speed of analysis. In scenarios such as clinical diagnosis or industrial process control, speed is of the essence. For example, blood gas analysis in an emergency room must deliver results within minutes, and furnace-front analysis in a metallurgical plant likewise demands rapid feedback to guide production adjustments.

Cost. Cost includes the purchase price of the instrument, routine operating costs (reagents, consumables, electricity), and the cost of personnel training. In teaching laboratories or small testing agencies with limited resources, cost is often the primary constraint.

Sample size and destructiveness. Some methods require relatively large samples (such as gravimetric analysis), while others need only microgram-level

samples (such as mass spectrometry). At the same time, some methods consume or destroy the sample (such as atomic absorption spectrometry), while others allow nondestructive testing (such as X-ray fluorescence spectrometry), which is of critical importance for applications such as the authentication of cultural relics.

Degree of automation and ease of operation. For laboratories processing large numbers of samples (such as clinical testing centers or environmental monitoring stations), highly automated methods can significantly raise throughput and reduce human operational errors.

It is worth pointing out that the criteria above often trade off against one another. For example, raising sensitivity often comes at the expense of speed, and pursuing high accuracy usually means higher cost. The selection of an analytical method is therefore not a search for the “best method,” but for “the method most suitable under the current constraints.”

From its earliest reliance on human taste to judge the properties of matter, to its present ability to probe the chemical composition of a single molecule or a single cell, analytical chemistry has come to depend more and more on systematic theoretical guidance. In the chapters that follow, when specific analytical methods are taught, they will also be analyzed and compared along the dimensions above, helping the reader gradually build the ability to evaluate and select analytical methods.

These selection criteria did not arise from nowhere; they are the distilled experience accumulated over several millennia of development of analytical chemistry. To truly understand them, it is necessary to review the past and present of this discipline.

Self-Check

Fill in the blanks: In each scenario below, what is the most critical criterion?

1. Blood gas analysis in an emergency room must produce results within minutes; the most critical criterion is _____.
2. Detecting ng/L-level trace pesticide residues in drinking water; the most critical criterion is _____.
3. Detecting drug components in forensic evidence identification; the most critical criterion is _____.

1.4 A Brief History of Analytical Chemistry

As the ancients said, history is a mirror through which one may understand the rise and fall of things. One of the more effective ways to understand a discipline is to study its history, and analytical chemistry is no exception.

Figure 1.3 presents, in the form of a timeline, the important milestones of analytical chemistry from 3000 BC to the present. The dedicated historical work on analytical chemistry is *The Development of the Methods of Analytical Chemistry*, published in 1960 by the Hungarian scientist Ferenc Szabadváry under the title *Az analitikai kémia módszereinek kialakulása*. In 1966 Gyula Svehla translated it into English under the title *History of Analytical Chemistry*. Published relatively early, this work covers the exploration of analytical chemistry from antiquity to the establishment of chromatographic theory in the twentieth century, spanning more than five thousand years. Since that book, although occasional chapters and review articles have appeared, no other systematic monograph devoted specifically to the history of analytical chemistry has been published. One important reason is that scientists have grown accustomed to incorporating the latest developments of analytical chemistry into the history of chemistry as a whole. This is

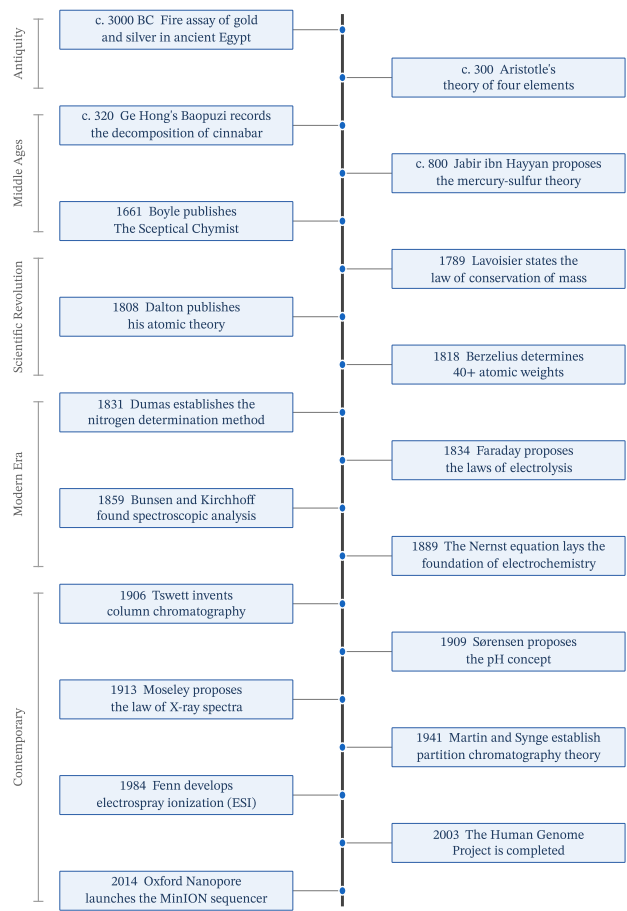


Figure 1.3: Timeline of the history of analytical chemistry.

no doubt intended to preserve the integrity of chemistry, but the distinctiveness of analytical chemistry has been gradually diluted.

In this section, too, we can only give a brief description of the history of analytical chemistry within a broad framework, so that the reader may gain a basic understanding of its overall development. Some topics will be taken up in the relevant chapters.

1.4.1 Analytical Chemistry in Antiquity

Ancient analytical chemistry is generally taken to cover the chemical practice of humanity from about 3000 BC to the fifth century AD. During this period, of course, there was no professional designation of chemistry, nor of analytical chemistry; there were only the ancients, who, on the basis of the laws of chemical change they had mastered, consciously applied those laws to daily production and life. The core of ancient analytical technique lay in solving practical problems, from coinage and water quality assessment to metal purification and mineral identification, spanning an extremely wide range. Fire assaying, densitometric measurement, and reagent testing constituted the three pillars of early analytical chemistry, while the philosophical speculations of element theory and alchemy supplied the logical core for the development of technique.

During the second and third centuries AD, alchemy was born in the Hellenistic world, with the core goal of transmuting base metals into gold and silver through experiment. The theoretical basis of alchemy can be traced back to Aristotle's¹ theory of four elements: since all things are composed of the four elements in certain proportions, changing the proportions of the elements would realize the transformation of matter. Hellenistic alchemists further proposed a

¹Some scholars today have begun to question whether Aristotle was a real historical person.

mercury-sulfur dualism, attempting to explain the differences among the properties of metals. Although these theories were shrouded in mystery, they gave birth to advances in separation techniques such as distillation and crystallization. For example, the *Leiden Papyrus* of the third century AD records methods of judging the purity of metals by heating, while the discovery that nitric acid dissolves silver (in the Roman period) promoted the innovation of gold-silver separation techniques. These practical explorations, though still confined to the realm of experience, laid the groundwork for quantitative analysis in medieval Europe and profoundly influenced the trust mechanism in the circulation of precious metals in trade between East and West.

The modern English word “alchemy” is said to derive from “Khemi,” the Egyptian word for “Egypt,” with the original meaning of “the Egyptian art.” According to historical records, after suppressing a minor revolt in Egypt, the Roman emperor Diocletian ordered the systematic collection of books dealing with the “marvelous art of making gold and silver” and their complete destruction. It is generally believed that he acted out of political considerations, fearing that the Egyptians might harbor ambitions of rebellion against Roman rule, emboldened by their accumulated wealth. This episode has much in common with the ancient Chinese saying, “to possess a treasure is to be guilty of a crime.”

Later, Arab scholars added the definite article “al” to the word, forming “alchemia,” a form that still clearly denotes “alchemy” even in modern usage. It is now widely held that the word “chemistry” evolved from “alchemy.” It is worth noting, however, that as early as AD 336, the astronomer and mathematician Firmicus (Julius Maternus Firmicus), who lived in the time of Constantine the Great, discussed the influence of the stars on human destiny in his writings, stating explicitly that “those born under the sign of Saturn should study chemistry.” This

is one of the earliest extant records of the word “chemistry” in Latin literature.

Ancient China also engaged in many early chemical practices. For example, Ge Hong, an alchemist of the Eastern Jin dynasty, wrote the Daoist classic *Baopuzi* (*The Master Who Embraces Simplicity*), which, apart from its rich Daoist thought, records a large number of practical activities involving the transformation of matter, providing precious material for the study of ancient chemical technique. In the *Baopuzi*, Ge Hong recorded many observed phenomena, some of the most famous of which include:

Historical Note

- “Cinnabar burned becomes mercury, and on repeated transformation reverts to cinnabar.” The cinnabar here is mercury(II) sulfide (HgS). On heating, it undergoes a decomposition reaction, yielding mercury (Hg) and sulfur (S); on cooling, the two can recombine.
- “Coat iron with azurite, and the iron turns red like copper; azurite upon meeting iron is transformed into copper.” The “azurite” here refers to a copper-containing mineral (such as azurite or malachite). When smeared on iron, a copper-colored substance is precipitated on the iron surface. This is in fact the displacement of copper ions from solution by iron, and is among the earliest observations of metal displacement reactions ever recorded.
- “Burn realgar together with sulfur, and where they are burned, purple quartz is formed.” The realgar here is tetrarsenic tetrasulfide (As_4S_4); heated together with sulfur, it may be converted into other arsenic sulfides.

Ge Hong’s records show that people at that time already had careful observations of chemical change, and were able to reveal transformation relationships among substances through practice. Such records are an important expression of the exploration of the properties of matter. For example, through repeated observation of the heating reaction of cinnabar, people came to understand its basic

composition and behavior, providing an empirical basis for later applications. In addition, the phenomenon recorded in the *Baopuzi* of “burning with fire, a purple-blue smoke rises” may be the characteristic color produced by certain metal salts heated in a flame. Although the ancients did not know the principle of spectral analysis, this observation of characteristic phenomena of matter testifies to their keen insight, and accumulated empirical knowledge that would later germinate into analytical methods.

Daoist elixir alchemy played a unique role in the development of ancient Chinese chemical technique. As a practice pursuing a specific goal, it impelled practitioners to perform numerous experiments in metal smelting, mineral decomposition, and substance synthesis, accumulating valuable experience in the transformation of matter and mastering certain purification and preparation techniques. However, the goal of elixir alchemy was mainly longevity and immortality; its theoretical framework was suffused with mysticism and lacked a systematic rational inquiry into the essence of chemical reactions. Its experimental methods relied mostly on experience and qualitative observation, lacking precise measurement and quantitative analysis. Therefore, although it accumulated a large body of practical experience, it failed to develop that experience into a systematic, scientific theoretical system of chemistry.

Even so, the extensive practical records and observations bequeathed by alchemy undoubtedly provided important historical material and inspiration for the formation and development of later chemical knowledge.

1.4.2 Analytical Chemistry in the Middle Ages

The Middle Ages generally refer to the period from the fifth to the sixteenth century. During this period, the techniques of analytical chemistry exerted a pro-

found influence on the history of chemistry; its theoretical foundations, experimental methods, and practical applications laid important foundations for the establishment of later chemical theory and for industrial progress.

In theory, Aristotle's doctrine of four elements and the mercury-sulfur theory of Jabir ibn Hayyan constituted the core framework of chemical thought at the time. The four-element doctrine held that matter is composed of four elements, earth, water, air, and fire, each representing a combination of two basic qualities, dryness or wetness and coldness or heat, providing a basis for understanding the nature of matter and its changes. Building on this, Jabir proposed the mercury-sulfur theory, holding that all metals are composed of mercury and sulfur combined in different proportions and forms, a view that drove alchemists to study in depth the properties and transformations of metals.

In experimental technique, improvements in distillation made it possible to prepare important chemicals such as nitric acid and sulfuric acid from minerals, providing key raw materials for later chemical experiments and industrial production. Jabir's recorded method for synthesizing nitric acid, by heating a mixture of green vitriol, saltpeter, and alum, testifies to the level of distillation technology of the time. Fire assaying, exploiting the collecting action of lead for precious metals, determined the gold and silver content through complex operations, and was widely applied to the control of coinage fineness and the assurance of metal quality. Quartation, adjusting the gold-silver ratio to 1 : 3 and exploiting the selective dissolution of silver by nitric acid to achieve efficient purification, significantly promoted the prosperity of silver mining in central Europe. In addition, the hydrometer was used to detect the adulteration of wine, and Venetian glass-makers used the silver nitrate color reaction to identify lead impurities. These techniques played an important role in improving product quality and regulating

market order.

The application of analytical techniques promoted the development of many industries. In mining and metallurgy, fire assaying and quartation raised the efficiency of metal smelting and the quality of finished products; in coinage, they helped establish an early quality supervision system and maintained economic stability; in handicrafts and trade, the related testing techniques safeguarded product reputation and promoted market prosperity.

Moreover, the analytical chemistry of the Middle Ages laid a solid foundation for the rise of modern chemistry. The four-element doctrine and the mercury-sulfur theory inspired later reflection on the composition of matter, providing an intellectual source for Boyle's concept of the element. The standardization of experimental methods such as distillation and fire assaying promoted the development of quantitative analysis. The experimental spirit upheld by the alchemists and their accumulated practical experience injected vitality into the birth of the modern chemical industry.

1.4.3 Analytical Chemistry in the Scientific Revolution

The sixteenth to eighteenth centuries were a critical period in the Scientific Revolution, and analytical chemistry achieved breakthroughs on many fronts in this stage, exerting far-reaching influence on the discipline of chemistry itself, on industry, and on social culture. During this period, the scientific way of thinking and method of research underwent a fundamental transformation, shifting from reliance on experience and conjecture toward an emphasis on experimental verification and quantitative analysis. This transformation infused new vitality into analytical chemistry, enabling it to develop from a mere technique of substance testing into an independent discipline with a systematic theory and scien-

tific method.

In 1661, Boyle (Robert Boyle) published the landmark work *The Sceptical Chymist*. The book systematically questioned and criticized the vague theories and practices of alchemy and iatrochemistry that then predominated. Boyle proposed that matter might be composed of certain “primitive and simple bodies,” components that remain unchanged in chemical transformations. He clearly distinguished these practical “elements” studied by chemists from the intangible “metaphysical elements” discussed by philosophers. This line of thought pointed clearly toward the fundamental units of matter, paving the way for the later birth of the scientific concept of the chemical element. Boyle’s work powerfully advanced the separation of chemistry from medicine, pushing it toward an independent natural science based on experiment, and pointed the way to the founding of modern chemistry.

By the middle of the eighteenth century, quantitative analysis techniques had already been widely applied in chemical research, setting the stage for a fundamental theoretical breakthrough. Lavoisier (Antoine-Laurent de Lavoisier) systematically reshaped the basic concepts of chemistry through a series of rigorous quantitative experiments. He proposed the operational definition that “an element is a substance that cannot be decomposed further,” connecting the concept of the element with the actual capabilities of chemical analysis at the time and thereby giving it a clear and practical meaning. In *Traité Élémentaire de Chimie*, published in 1789, he listed 33 substances as elements, including oxygen, hydrogen, and nitrogen; although some were later shown to be compounds, this was nevertheless the first systematic list of elements based on the new concept, marking the beginning of the scientific recognition of the elements.

Lavoisier’s greatest contribution was his explicit formulation and systematic

application of the “principle of the indestructibility of matter” (the law of conservation of mass). By repeating and improving earlier sealed-vessel combustion experiments (such as Boyle’s) and, with collaborators, decomposing and synthesizing water, he finally advanced the oxygen theory of combustion and overthrew the long-dominant phlogiston theory. He showed that combustion is the combination of matter with oxygen and that the total mass remains unchanged before and after the reaction. In his writings he declared: “... ..nothing is created, either in natural processes or in artificial operations we can state it as a principle: in all operations before and after, the quantity of matter is the same; only changes and transformations occur.” This entire body of theory, including a clear definition of the element, the principle of conservation of mass, and the oxygen theory, not only explained scientifically the core phenomena of combustion but also placed chemistry on a solid quantitative foundation, completely redirecting the course of chemical research and laying the cornerstone of its emergence as a modern science.

Dalton’s (John Dalton) atomic theory laid the crucial microscopic foundation of chemistry. His ideas were first presented in a lecture in 1803, and his systematic atomic theory was formally published in 1808 in *A New System of Chemical Philosophy*. Dalton held that elements consist of indivisible atoms of definite weights, and that the essence of chemical reactions is the recombination of atoms. In studying the oxides of carbon and nitrogen, Dalton found that the masses of one element combining with a fixed mass of another element show simple integer ratios (for example 1:2). From this he summarized the Law of Multiple Proportions: when two elements can form more than one compound, the masses of element B combining with the same mass of element A in these compounds stand in the ratio of simple integers. The proposal of this law had far-reaching significance.

It not only provided a quantitative basis for mineral component analysis and the derivation of chemical formulas, but also powerfully supported and completed the fledgling atomic theory from experiment, showing that the composition of compounds indeed follows fixed ratios at the atomic level. This work greatly advanced the development of stoichiometry, and was a key step in chemistry's transition from accumulated experience to a systematic theoretical system.

In the early nineteenth century (especially 1807-1818), Berzelius (Jöns Jacob Berzelius) accomplished a systematic undertaking that laid the foundation of modern chemistry: the determination of the atomic weights of a great number of elements. He chose oxygen (set to 100) rather than hydrogen as the standard, mainly because much of his work was based on the analysis of oxides. Under the harsh conditions of the time, when commercial chemicals were extremely scarce and most reagents had to be prepared by hand, he spent years measuring the atomic weights of more than forty elements, most of which were already close to modern values. This work was far more than the provision of a standardized set of data. It made Dalton's atomic theory and the laws of stoichiometry truly usable for precise calculation, and thereby drove the paradigm shift of chemical analysis from qualitative to quantitative. The quantitative analysis procedures that Berzelius established and perfected (often involving the conversion of metallic components into stable oxides for weighing) significantly improved the precision and comparability of component analysis of complex samples such as ores and minerals. The precise atomic weights and systematic quantitative analytical methods he established provided a practical foundation for stoichiometry. This scientific method based on quantitative relationships later became the universal and indispensable tool for component analysis and quality control in all materials science and industrial production.

1.4.4 Analytical Chemistry in the Modern Era

In the early nineteenth century, with the rise of organic chemistry, the need for precise elemental analysis of organic compounds became increasingly urgent. The methods of the time, however, had obvious limitations; especially for the analysis of nitrogen-containing compounds, the lack of a unified and reliable standard procedure made the results of different laboratories differ markedly.

In 1831, the French chemist Dumas (Jean-Baptiste André Dumas), building on the work of his predecessors, proposed a new method for nitrogen determination that became a milestone in the field. The method achieved, for the first time, the standardization of the analysis of nitrogen in organic compounds, and its core principle is still in use today. The Dumas method is based on high-temperature decomposition of the nitrogen-containing organic compound and quantitative analysis of the gaseous products. The procedure is as follows: the sample is mixed with copper(II) oxide and heated in a tube furnace at about $600 \sim 800\text{ }^{\circ}\text{C}$. Carbon and hydrogen in the sample are converted into carbon dioxide and water, respectively, while nitrogen is converted into nitrogen oxides. A key innovation of the Dumas method is the use of a stream of carbon dioxide, generated by heating lead carbonate, to sweep and carry all the reaction gases. The gas stream then passes through a section packed with copper that has been activated by reduction in hydrogen, where the nitrogen oxides are quantitatively reduced to nitrogen gas. Finally, the mixed gas is introduced into a eudiometer filled with concentrated potassium hydroxide solution. The carbon dioxide is absorbed by the alkali, the water condenses, and the volume of the remaining gas is the volume of nitrogen. At standard conditions, from the ideal gas equation of state and the measured volume of nitrogen, the amount of substance can be calculated, and from it the nitrogen content of the sample can be deduced.

The contribution of the method to the standardization of the analysis of nitrogen-containing organic compounds is mainly threefold. First, it solved the earlier problem that methods differed and results were not comparable. Second, by standardizing the key operations (uniform use of copper(II) oxide and activated reduced copper, control of the reaction temperature, and regulated measurement of gas volumes), it ensured the repeatability of results. Third, it established, for the first time, a complete quantitative determination system based on gas volume measurement and physical laws. The establishment of the Dumas method greatly promoted the determination of molecular formulas and the elucidation of structures of nitrogen-containing compounds such as alkaloids, providing vital technical support for the flourishing development of organic chemistry in the nineteenth century.

In the same period, the German chemist Liebig (Justus von Liebig) made a landmark contribution to organic elemental analysis. He made key improvements to Berzelius's organic combustion analysis method, publishing his improved method and apparatus in 1837. Liebig's improvements consisted mainly of three measures: first, he designed a combustion tube furnace with separately adjustable heating zones to better control the decomposition process; second, he changed the measurement of carbon dioxide from the volumetric method to the gravimetric method, determining the carbon content by weighing the increase in mass of an absorption bulb containing alkali, which improved accuracy; and third, he systematically used anhydrous calcium chloride to absorb the water formed, in order to determine the hydrogen content. These improvements markedly increased the precision and reliability of the determination of carbon and hydrogen in organic compounds, and the basic principles of his method remain the basis of modern elemental analyzers.

Interestingly, Liebig and Berzelius engaged in heated arguments over details of the method, such as the best material for connecting tubes (cork vs. rubber tubing) and whether the air used for sweeping should be purified in advance. These debates themselves drove the refinement of analytical technique. In 1837, Liebig published *Anleitung zur Analyse organischer Körper (Instructions for the Analysis of Organic Substances)*, systematically expounding his methods and theories of organic elemental analysis. The work explained in detail the principles, operating steps, apparatus, and calculations, and discussed potential sources of error, providing a crucially important textual basis for the standardization and normalization of organic analytical methods, and greatly accelerating the rapid development of nineteenth-century organic chemistry.

In electrochemistry, Faraday (Michael Faraday) proposed the famous laws of electrolysis in 1834, laying a solid theoretical foundation for electroanalytical chemistry. His work continued and deepened the earlier observations of Wollaston and others. Faraday's laws of electrolysis consist of two closely related regularities. The first law states that during electrolysis, the mass m of substance deposited (or dissolved) at an electrode is proportional to the total quantity of electricity Q passing through the cell, that is, $m = kQ = kIt$, where k is the electrochemical equivalent of the substance, I the current, and t the time of passage. The second law states that when the same quantity of electricity passes through different electrolytes, the masses of substances deposited at the electrodes are proportional to their chemical equivalents (the ratio M/n of molar mass M to ionic charge number n). In other words, the electrochemical equivalents k of different substances are proportional to their chemical equivalents M/n . Combining the two laws yields $m = (MQ)/(Fn)$, where the proportionality constant F is the Faraday constant, the quantity of electricity required to deposit 1 equivalent of

substance (that is, M/n grams). It was later measured precisely as about 96485 coulombs per mole (C/mol). These two laws not only provided the direct theoretical basis and method of calculation for electrogravimetry, the determination of the content of a metal in solution by weighing the deposit on an electrode, but also established fundamentally the quantitative relationship between quantity of electricity and quantity of chemical reaction, making possible the precise determination of composition by electrical means. They constitute a milestone in the history of analytical chemistry.

In the middle of the nineteenth century, as chemical research and industrial needs grew, the systematization and standardization of analytical methods became increasingly urgent. The German chemist Fresenius (Carl Remigius Fresenius) made foundational contributions in this period. In 1841 he published *Anleitung zur qualitativen chemischen Analyse (Introduction to Qualitative Chemical Analysis)*. The book achieved immediate and great success; by 1852 it had reached a seventh edition and had been translated into many languages. In this work, Fresenius proposed a clearly organized “systematic qualitative analysis of cations”², dividing the most important cations into six groups according to the characteristic reactions of their metals (oxides), and laying out a logically rigorous step-by-step analysis scheme. This scheme successfully remedied the defect of earlier works, which was cluttered with information and in which beginners easily lost their way. Its orderliness and practicality are such that teaching and practice of modern qualitative analysis still basically follow Fresenius’s system.

Subsequently, in 1846, he published *Anleitung zur quantitativen Analyse (Instructions for Quantitative Analysis)*. This book provides extremely detailed op-

²Early Chinese analytical chemistry textbooks included systematic qualitative analysis of ions, but this content was later deleted in most. This book focuses on gravimetric and volumetric analysis, which remain the core of chemical quantitative analysis.

erating instructions for gravimetric analysis, covering the whole process of precipitation, filtration, washing, drying or ignition, and calculation, and discusses methods of separating many elements. It should be noted, however, that at the time of its publication, volumetric analysis (titrimetry) had not yet been fully accepted by the mainstream analytical chemistry community. Fresenius himself admitted in the preface to the book that although the volume of a solution was sometimes measured in industrial analysis, it was “difficult to obtain accurate results with the required apparatus ... and therefore in important analyses the balance is preferred.” The popularization of volumetric analysis and its entry into standard textbooks came only after the 1860s. Apart from textbooks, Fresenius also founded in 1862 one of the most influential early professional journals in analytical chemistry, the *Zeitschrift für analytische Chemie*, marking the maturation of analytical chemistry as an independent discipline. Through his classic textbooks and the founding of a professional journal, Fresenius greatly promoted and accomplished the normalization and systematization of the education and practice of analytical chemistry, laying a solid foundation for the independence and development of the discipline.

In the same period, advances in physics and technology gave birth to new analytical paradigms. In 1859, Kirchhoff (Gustav Kirchhoff) and Bunsen (Robert Bunsen) cooperated to lay the foundations of spectrochemical analysis. Kirchhoff was the principal inventor of the spectroscope and elucidated its theoretical principles, while Bunsen provided an ideal excitation source with his invention of the flameless gas burner (the Bunsen burner). Using a prism to disperse the light emitted by substances excited in a flame, they observed that each element produces a unique set of invariant bright spectral lines, like the “fingerprint” of the element. This marked the birth of spectroscopic analysis, effecting the impor-

tant transition from wet chemical analysis based on chemical reactions to instrumental analysis based on physical measurements. The technique immediately displayed tremendous power. With its extremely high sensitivity (capable of detecting traces of substances) and absolute specificity, Kirchhoff and Bunsen discovered the new elements cesium and rubidium in 1860 and 1861 by analyzing the spectrum of mineral water from Dürkheim. Thereafter, spectroscopic analysis was rapidly applied in geological prospecting, ore identification, industrial quality control, and soon became indispensable in forensic science and toxicology. The founding of spectroscopic analysis was a fundamental revolution in the history of analytical chemistry: it opened a wholly new route to identification and detection that did not depend on traditional chemical separation and reactions.

In the late nineteenth century, solution chemistry and electrochemistry research advanced significantly. In 1883, the Swedish chemist Arrhenius (Svante Arrhenius) proposed the epoch-making ionic theory in his doctoral dissertation. The theory holds that electrolytes in solution spontaneously dissociate into positively and negatively charged ions, and that the degree of dissociation increases with dilution. This theory successfully explained a series of physicochemical properties of solutions, such as electrical conductivity and osmotic pressure anomalies. Arrhenius's ionic theory provided a key theoretical basis for understanding the microscopic mechanisms of changes in solution conductivity, thereby indirectly supporting the development of conductometric titration. In fact, conductometric titration was first described by Behrend (Robert Behrend) in 1883. This method determines the endpoint of a titration by precisely measuring the change in the electrical conductivity of the solution during titration. Conductometric titration is especially suitable for systems that are difficult to handle by traditional indicator methods, such as the titration of weak acids with weak bases, or solutions

that are themselves colored or turbid. It has therefore found important applications in environmental monitoring, the pharmaceutical industry, and component analysis in chemical production.

In 1889, Nernst (Walther Nernst) proposed the famous Nernst equation. This equation, for the first time, quantitatively described the relationship between electrode potential and the activities of the ions participating in the electrode reaction and the temperature (strictly, activities should be used; in dilute solutions, concentrations are an acceptable approximation), providing a powerful mathematical tool for the study of electrochemical equilibrium. His work built on the research results of van't Hoff (Jacobus H. van't Hoff), Arrhenius, and others. The Nernst equation laid the foundation for understanding and calculating redox titration curves from theory. In fact, in the theorization of titration analysis, Ostwald (Friedrich Wilhelm Ostwald) was the first, in 1894, to systematically apply the law of mass action and the ionic theory to the titration process and to plot titration curves, thereby providing a theoretical framework for explaining the potential (or pH) jump near the endpoint. In modern analytical chemistry, therefore, constructing titration curves based on the Nernst equation and performing quantitative analysis with them has become standard content in chapters on redox titration.

The precision of classical quantitative analysis reached a peak at the end of the nineteenth century. In 1894, the British physicist Rayleigh (Third Baron Rayleigh), using gravimetric analysis to determine the density of nitrogen, discovered a tiny difference of seven parts in ten thousand: nitrogen separated from air was slightly heavier than nitrogen prepared by chemical means. After three rounds of repeated experiments and cross-verification, he ruled out experimental error and proposed the hypothesis that “there may be an unknown impurity gas

in atmospheric nitrogen.” This discovery attracted the attention of the chemist Ramsay (William Ramsay). Combining the predictions of Mendeleev’s (Dmitri Mendeleev) periodic table, Ramsay liquefied air, progressively removed the oxygen and nitrogen, and determined the density of the remaining gas by gravimetric analysis to be 1.97 g/L, confirming by spectroscopic analysis that it was an inert gas. On August 13, 1895, the two jointly announced the discovery of the new element “argon” (Argon, from the Greek “lazy”). This discovery not only verified the precision of gravimetric analysis in determining gas densities, but also embodied the interdisciplinary synergy of gravimetric analysis, spectroscopic analysis, and theoretical prediction: it filled a gap in Mendeleev’s periodic table, ended the traditional belief that the atmosphere contained only oxygen and nitrogen, and opened a new era of research on the noble gases.

1.4.5 Contemporary Analytical Chemistry

As an important branch of chemical science, analytical chemistry has undergone profound changes since the twentieth century: it has evolved from classical analytical methods to modern instrumental analysis, and is now moving toward a new stage of interdisciplinary integration and intelligence. This development has not only greatly advanced science and technology, but has also widely influenced industrial production and many aspects of social life.

The Transition from Classical Analysis to Instrumental Analysis

At the beginning of the twentieth century, traditional methods grew increasingly inadequate for the challenge of separating complex mixtures. In 1906, the Russian botanist Tswett (Mikhail Tswett) invented column chromatography, providing a pioneering solution. In his classic experiment, chalk (calcium carbon-

ate) powder was packed into a glass column as the stationary phase, a petroleum ether extract of chlorophyll was applied, and pure petroleum ether was then used as the mobile phase for elution, successfully separating the plant pigments in the column. Tswett named the method “chromatography” (meaning “color writing”). This important invention, however, did not attract sufficient attention in the chemical community for several decades.

Chromatography truly acquired a solid theoretical foundation and wide dissemination thanks to the work of the British scientists Martin (Archer Martin) and Synge (Richard Synge). In 1941 they published the foundational theory of partition chromatography, pointing out that the essence of chromatographic separation is a continuous partition process of the solute between two immiscible liquid phases (the stationary phase and the mobile phase), based on differences in partition coefficients. Building on this theory, the two developed paper chromatography in 1944, using the water adsorbed on filter paper fibers as the stationary phase and an organic solvent as the mobile phase, so that the components of a sample migrate along the paper and form separated spots. This method was simple, efficient, and rapidly adopted. The theory and practice of Martin and Synge paved the way for the development of modern chromatographic techniques, including gas chromatography and high-performance liquid chromatography, and constitute a milestone in the history of analytical chemistry.

At the same time, the precise measurement and expression of the acidity or basicity of solutions became an important topic in analytical chemistry, especially in biochemistry. In 1909, the Danish chemist Sørensen (Søren Peder Lauritz Sørensen), working at the Carlsberg Laboratory on enzymatic reactions, first proposed the pH concept for the sake of a more convenient expression of hydrogen ion concentration. He defined it as “the negative common logarithm of the

hydrogen ion concentration" ($\text{pH} = -\log [\text{H}^+]$). This concise scale greatly facilitated the quantification of acidity and basicity, and had a profound influence on research in biochemistry and fermentation processes.

Early pH determination relied mainly on colorimetric methods, that is, the use of a series of indicators that change color over different pH ranges. Sørensen himself evaluated a large number of synthetic and natural indicators. Reliable colorimetric methods depended crucially on the use of buffer solutions, a concept proposed by Szily (Pal Szily) around 1904.

In electrochemical measurement, in 1909 the German chemist Haber (Fritz Haber) and Klemensiewicz (Zygmunt Klemensiewicz) pioneered the glass electrode. The earliest glass electrodes, however, suffered from high asymmetry potentials and poor stability. After nearly two decades of continuous improvement (particularly the overcoming of the "alkaline error"), glass electrode technology matured in the late 1920s and early 1930s. In 1935, Beckman (Arnold Orville Beckman) introduced the first commercial pH meter, making rapid and accurate potentiometric pH measurement widely available in laboratories and industrial production.

In the field of spectroscopy, physics in the early twentieth century was on the eve of revolutionary breakthroughs. At the time, the ordering of several pairs of elements in Mendeleev's periodic table (such as cobalt/nickel, tellurium/iodine) contradicted their atomic weights, raising doubts about the fundamental basis for ordering the elements. In 1913, the British physicist Moseley (Henry Gwyn Jeffreys Moseley), through systematic study of the X-ray spectra of many elements, proposed the famous Moseley's law, laying the theoretical foundation of X-ray spectrochemical analysis.

Moseley found that the square root of the frequency (ν) of the character-

istic X -rays emitted by an element is linearly related to its atomic number (Z): $\sqrt{\nu} = a(Z - \sigma)$, where a is a proportionality constant and σ is the shielding constant. This law connected the X -ray spectrum directly with the positive charge number of the atomic nucleus, namely the atomic number (Z), rather than with the atomic weight. This discovery proved irrefutably that the atomic number is a more fundamental property of an element than the atomic weight, and established its decisive position in the periodic table. It directly corrected the ordering of cobalt and nickel, argon and potassium, and tellurium and iodine in Mendeleev's periodic table. More importantly, Moseley's law enabled him to predict the existence of elements of specific atomic numbers not yet discovered at the time (such as 43, 61, 72, 75), pointing the way to the later discovery of new elements, and powerfully supported the nascent theory of atomic structure.

Moseley's research had a profound influence on the development of quantum theory and on the discovery of several new elements (such as technetium and rhenium). By the 1940s, with the marked improvement of X -ray tube technology and detector performance, X -ray fluorescence spectrometry based on his principle gradually left the laboratory and was widely applied in geological prospecting, the metallurgical industry, and cultural relic identification, becoming an important nondestructive elemental analysis technique.³

The Golden Age of Instrumental Analysis

In the 1930s, mass spectrometry achieved a key breakthrough in analytical chemistry. The American scientist Nier (Alfred O. C. Nier) made a series of fundamental improvements to the magnetic sector mass spectrometer, a milestone in the history of the technique. Nier built a small, compact, and efficient mass

³Tragically, Moseley was killed in the First World War in 1915, at the age of only 27.

spectrometer. Its core innovation was the use of a semicircular (180°) uniform magnetic field to achieve precise focusing of the ion beam, markedly improving the resolution and sensitivity of the instrument. These improvements laid a solid technical foundation for the precise measurement of isotopic abundances, enabling mass spectrometry to be widely applied in geochemistry, cosmochemistry, nuclear physics, geochronology, and other frontier fields. Using his improved instrument, Nier determined for the first time the natural abundances of uranium isotopes (such as uranium-235 and uranium-238). This achievement provided the scientific data vital to the separation of uranium-235 required by the “Manhattan Project” during the Second World War. Nier’s work marked the transformation of the mass spectrometer from a bulky, complex laboratory device into a practical analytical tool, and was a key turning point in the history of mass spectrometry.

In the 1980s, mass spectrometry faced a fundamental challenge in analyzing polar, thermally unstable biological macromolecules such as proteins, because these molecules are difficult to convert into gas-phase ions without decomposition. In 1984, the American scientist Fenn (John Fenn) and his collaborator Yamashita (Masamichi Yamashita) successfully developed electrospray ionization (ESI). This technique forms charged microdroplets from the analyte in solution under a high-voltage electric field and, after a desolvation process, generates gas-phase ions, thereby achieving efficient ionization and mass analysis of intact proteins and other large molecules. Almost at the same time, laser desorption techniques also achieved a breakthrough. The Japanese scientist Tanaka (Koichi Tanaka) discovered as early as 1985 that using ultrafine metal powder mixed with glycerol as a matrix could achieve soft laser desorption ionization of proteins, and first publicly reported the technique at an academic conference in 1987. Nearly simultaneously, the German scientists Karas (Michael Karas) and Hillenkamp

(Franz Hillenkamp) formally published matrix-assisted laser desorption/ionization (MALDI) in 1987, using organic acid matrices to assist ionization. These two breakthrough technologies (ESI and laser desorption) enabled mass spectrometry to analyze proteins, peptides, nucleic acids, polysaccharides, and other biological macromolecules directly and rapidly, with unprecedented gains in sensitivity and mass range. They drove the revolutionary development of proteomics and modern life science research.

In chromatography and mass spectrometry, a pioneering breakthrough occurred in 1955. Researchers Gohlke (Roland S. Gohlke) and McLafferty (Fred W. McLafferty) of the spectroscopy laboratory of the Dow Chemical Company successfully coupled a homemade gas chromatograph with a research time-of-flight mass spectrometer (TOF MS). Around December 1955, at the Bendix research laboratory in Southfield, Michigan, they used this unique GC/MS instrument to achieve, for the first time, rapid on-line mass analysis of gas chromatographic eluates. The instrument generated mass spectra at a rate of 10 kHz and displayed them on an oscilloscope, allowing them to identify visually and in real time compounds such as methanol, acetone, benzene, toluene, and carbon tetrachloride as they were separated. The main motivations driving the research included identifying the trace components separated by GC, exploring the capabilities of the new fast mass spectrometer, and foreseeing the enormous potential of GC-MS coupling. This pioneering work marked the beginning of a new era of high-precision analysis of complex mixtures, and is the cornerstone of modern chromatography-mass spectrometry hyphenated techniques.

Against the background of growing demands for microstructural research in materials science and physics, in 1951 the German scientist Müller (Erwin Wilhelm Müller) invented the field ion microscope (FIM), a technique of pioneering

significance for surface analysis. FIM is based on the principle of field ionization: a needle-shaped sample is placed in an ultrahigh vacuum and a high voltage is applied between the sample and a fluorescent screen, creating an extremely strong electric field. Under this field, inert gas molecules are ionized and accelerated to strike the fluorescent screen, revealing the spatial distribution of atoms on the sample surface, and achieving for the first time the direct observation of atomic structure.

In 1981, the Swiss scientists Binnig (Gerd Binnig) and Rohrer (Heinrich Rohrer) successfully developed the scanning tunneling microscope (STM) at the IBM Zurich Research Laboratory. Based on the quantum tunneling effect, the technique monitors the tunneling current (typically in the range of pA to nA) flowing across a nanometer-scale gap between a sharp metal tip and a conductive sample surface, achieving for the first time real-time, high-resolution imaging of the atomic structure of material surfaces. This breakthrough achievement earned Binnig and Rohrer the 1986 Nobel Prize in Physics, shared with Ruska (Ernst Ruska), the inventor of the electron microscope. The STM has atomic-level resolution (up to 0.1 nm laterally and 0.01 nm vertically), operates stably in vacuum, air, liquids, and other environments, and has therefore become an indispensable core tool in surface science and nanotechnology research.

At the end of the twentieth century and the beginning of the twenty-first, analytical chemistry made marked progress through interdisciplinary integration. In 1999, the American scientist Quake (Stephen Quake) developed microfluidic chip technology, bringing a revolutionary breakthrough to single-cell analysis. Based on microelectromechanical systems (MEMS) fabrication processes, the technology can precisely manipulate fluids at the micro- and nanoscale and construct highly integrated microfluidic networks. In recent years, microfluidic chip

technology has been further integrated with biomedicine, materials science, and many other fields. In disease diagnosis, for example, microfluidic chips enable rapid, high-throughput detection of biomarkers, fully demonstrating their broad application prospects in precision medicine and high-throughput analysis.

In single-cell analysis, the single-cell RNA sequencing technology (scRNA-seq) developed in 2015 by the research group of the American scientist Ren (Bing Ren) is a landmark breakthrough, providing a key tool for dissecting cellular heterogeneity. The technology encompasses numerous precise steps, including single-cell isolation, barcode labeling, RNA reverse transcription and amplification, and high-throughput library construction, and can reveal the gene expression profiles of individual cells at the whole-transcriptome level. Since its introduction, single-cell RNA sequencing has greatly advanced research on the heterogeneity of cell types, states, and functions in developmental biology, tumor immunology, and neuroscience, and has become an indispensable core technology in modern life science research.

In nanopore sequencing, the MinION sequencer that Oxford Nanopore Technologies introduced in 2014 constitutes a landmark breakthrough in single-molecule sensing technology. Based on the intersection of biophysics and electrical principles, the technology achieves real-time, long-read sequencing of single-stranded DNA molecules through protein nanopores, bringing revolutionary progress to the field of gene sequencing. In recent years, nanopore sequencing has continued to deepen interdisciplinary integration; for example, its close combination with bioinformatics methods has markedly improved the accuracy and efficiency of sequencing data analysis, expanding its application prospects in pathogen detection, environmental genomics, and real-time monitoring.

The Modernization of Analytical Chemistry in China

Since the 1940s, analytical chemistry in China has achieved a series of landmark breakthroughs. In mass spectrometry, Chinese scientists successfully carried out precise analysis of uranium isotopes, providing key data support for the “Two Bombs, One Satellite” program; early exploration of chromatography-mass spectrometry hyphenated techniques also laid the foundation for the high-precision analysis of complex mixtures. In electroanalytical chemistry, Academician Wang Erkang successfully developed China’s first pulse polarograph, and Academician Yu Ruqin initiated the research direction of chemical biosensors, promoting disciplinary development and independent instrument research and development in the field. Academician Fang Zhaolun made major breakthroughs in flow injection analysis and microfluidic chip technology; the microfluidic technology he led has been hailed by the international community as “the chip of the poor man’s wisdom.” In spectroscopy, Chinese scientists achieved a series of important results in the analysis of rare earth elements and the detection of ultrapure substances.

In recent years, high-end domestic analytical instruments have continued to emerge; for example, the successful development of Fourier transform mass spectrometers and 600 MHz nuclear magnetic resonance spectrometers marks the leap of China’s analytical chemistry instruments from importation and imitation to independent innovation. These achievements have not only strongly supported major national strategic needs, but have also pushed China’s analytical chemistry research to the international frontier, and laid a solid foundation for the cross-domain integration and development of new technologies such as artificial intelligence and quantum sensing.

Prospects for Future Trends

As an important branch of chemical science, analytical chemistry, driven by the dual forces of rapid scientific and technological development and sustained social demand, is showing a multidimensional, cross-disciplinary trend of future development. At the technical level, analytical chemistry is continuously moving toward higher sensitivity and resolution. The integration of emerging materials and technologies such as nanotechnology, microfluidics, quantum dots, and two-dimensional materials is pushing analytical instruments to achieve precise detection of extremely small amounts of substances, gradually reaching the single-molecule and single-cell level.

The combination of multiple analytical techniques is gradually becoming the mainstream strategy. The efficient integration of chromatography, spectroscopy, and mass spectrometry gives full play to the advantages of each, achieving more comprehensive and more precise multidimensional analysis of complex systems. Intelligence and automation are also profoundly changing the research paradigm of analytical chemistry. The deep integration of analytical instruments with computers and microprocessor systems has realized full-process automation from sample preparation and experimental operations to data analysis, significantly improving experimental efficiency and the reliability of results.

Portable and on-site real-time analysis capabilities have seen important breakthroughs. Advances in micro- and nanofabrication and integration technologies are making analytical devices increasingly miniaturized and modular, and capable of being embedded in smartphones and other mobile terminals, enabling rapid detection and real-time monitoring in the field, in clinical settings, and at production sites, greatly expanding the practical scope of analytical chemistry.

In application fields, analytical chemistry will play an even more central role in key scenarios. In medicine and health, analytical chemistry techniques will contribute to early disease diagnosis, personalized treatment, and drug development. For example, real-time in vivo monitoring technology based on biomolecular recognition can provide powerful tools for personalized health management and disease early warning. In environmental monitoring, highly sensitive analytical means can precisely detect and track pollutants, providing a scientific basis for environmental governance and policy formulation. In materials science, analytical chemistry methods can reveal the structure-performance relationships of nanomaterials, bio-based materials, and high-performance composites, promoting the innovation and application of new materials. In food safety, analytical techniques can more precisely detect harmful substances, additives, and nutritional components, providing support for safeguarding food safety and public health.

From the perspective of green and sustainable development, analytical chemistry is actively moving toward an environmentally friendly future. Green analytical chemistry will become an important research direction, developing environmentally friendly analytical methods that reduce reagent use and waste generation, and lower the negative impact on the environment. At the same time, the increasing use of renewable resources and bio-based materials in analytical devices and sensors not only reduces dependence on nonrenewable resources, but also pushes the entire field toward a more sustainable direction, achieving a win-win situation for economic and environmental benefits and contributing to the global sustainable development goals.

From antiquity to the future, analytical chemistry has always responded to the practical needs of human society. The fields presented below are vivid con-

temporary practices of the discipline.

1.5 Application Fields of Analytical Chemistry

The applications of analytical chemistry are extremely broad. According to the field of application, it can be divided into pharmaceutical analysis, biomedical analysis, food analysis, industrial analysis, forensic analysis, clinical analysis, environmental analysis, and so on. In the following, we illustrate the key role of analytical chemistry in each field through several typical cases.

In pharmaceutical analysis, analytical chemistry plays the crucial role of “quality gatekeeper,” comprehensively safeguarding drug quality, supporting new drug development, and achieving precise resolution of chiral drugs. The “thalidomide” incident of the 1950s and 1960s fully demonstrates the key role of analytical chemistry in this process. Thalidomide, sold under the brand name “Contergan,” was widely used in Europe because it effectively relieved pregnancy-related nausea and vomiting. However, about 10,000 cases of newborn deformities subsequently shocked the world. Through in-depth research, scientists, with the aid of advanced analytical techniques, discovered that thalidomide is a racemic mixture composed of two enantiomers, dextrorotatory and levorotatory: the dextrorotatory enantiomer has a sedative effect, while the levorotatory enantiomer is teratogenic. This discovery not only revealed the irreplaceable role of analytical chemistry in drug quality control and structural identification, but also highlighted the importance of precisely identifying differences in the microstructures of drugs.

It should be noted that the story of thalidomide did not end with its severe adverse effects. Its sedative effect kept scientists from abandoning it. Scientists conducted long-term research on thalidomide and discovered, almost by chance, that it could be used to treat multiple myeloma. In 2006, the U.S. FDA approved

it for the treatment of multiple myeloma, providing a new therapeutic option for many patients. In this process, analytical chemistry played a central role, participating throughout the redevelopment and application of thalidomide, from precise analysis of drug components and identification of isomer risk to supporting drug modification and the exploration of new therapeutic efficacy.

In biomedical analysis, analytical chemistry often plays the role of an “acceleration engine.” In 1990, the “Human Genome Project”, led by the U.S. Department of Energy, was officially launched, with the core goal of determining the complete DNA sequence of the 3 billion base pairs of the human genome. This project of the century, ranked alongside the “Manhattan Project” and the “Apollo Program,” initially employed Sanger sequencing, which achieves base recognition through dideoxy chain termination reactions and fluorescence labeling. Although this method has an accuracy as high as 99.99% and is hailed as the “gold standard” of DNA sequencing, sequencing a single human genome cost about \$500 million and was expected to take 15 years. A key turning point came in 1992, when the team of Mathies (Richard A. Mathies) at the California Institute of Technology developed a microfluidic chip high-throughput sequencing system that integrated sample processing, electrophoretic separation, and fluorescence detection modules, increasing throughput by more than 100 times. On April 14, 2003, the Human Genome Project was completed two years ahead of schedule, revealing 97% of the human genome sequence; China, as the sixth participating country, contributed 1% of the sequencing work. The core roles of analytical chemistry are reflected in: (1) developing four-color fluorescently labeled dideoxynucleoside triphosphates to achieve automated sequencing; (2) optimizing enzymatic reaction systems to support the development of next-generation sequencing; and (3) integrating photolithography, electrophoresis, and molecu-

lar recognition technologies through microfluidic chips to promote the scaling of sequencing.

In food analysis, analytical chemistry is an indispensable technical means for safeguarding food safety. Its core role lies in accurately identifying food components, detecting contaminants, and monitoring production processes, thereby ensuring food quality and consumer health. The rancidity of edible oils is a case in point. This common food safety problem can be effectively monitored by analytical chemistry methods. During storage, oils are susceptible to oxidation induced by light, heat, oxygen, and other factors, producing harmful substances such as aldehydes, ketones, and peroxides, which not only degrade flavor but may also endanger human health. Traditional detection methods, such as peroxide value and acid value determinations, reflect the degree of oil oxidation only partially, and are limited in specificity, sensitivity, and early warning. Modern analytical means, including gas chromatography-mass spectrometry (GC-MS) and high-performance liquid chromatography (HPLC), can precisely identify and quantify trace harmful oxidation products, achieving a more comprehensive assessment of oil quality. By establishing strict analytical standards and procedures, analytical chemistry provides reliable data support for food manufacturers and regulatory agencies, helping to realize whole-process quality control from raw materials to finished products.

In industrial analysis, analytical chemistry plays a key role, especially in achieving high-precision manufacturing and product quality control. Semiconductor chip manufacturing is a typical example: the process is technology-intensive and complex, and places extremely high demands on material purity. As the core substrate of chips, the purity of silicon wafers directly affects chip performance and reliability. Using highly sensitive analytical techniques such as

inductively coupled plasma mass spectrometry (ICP-MS), trace metal impurities in silicon wafers, such as boron, phosphorus, and copper, can be accurately detected even at concentrations as low as parts per billion (ppb). Once impurities are found to exceed the allowable range, the production process can be adjusted in time, for example by optimizing raw material purification methods or strengthening cleanroom control of the production environment, thereby safeguarding the high purity of silicon wafers and providing a reliable material basis for chip manufacturing. The application of analytical chemistry in this process not only supports the technological progress of the semiconductor industry, but also provides a solid guarantee for breaking through process bottlenecks and promoting innovative development of the industry.

In forensic analysis, analytical chemistry serves as a key technical support, providing scientific evidence and decisive clues for the solving of many complex criminal cases. The arson case in California in 2003 is an example. A large commercial warehouse was completely destroyed by fire. The scene was heavily damaged, and the high temperature destroyed most conventional physical evidence, bringing the investigation to a standstill. The investigators, however, did not give up. Through systematic screening of the residual ash, they found some tiny incompletely burned clumps of material. These items of evidence were sent to a professional laboratory, where analysts performed detailed testing using gas chromatography-mass spectrometry (GC-MS). In this method, gas chromatography first achieves effective separation of compounds according to their partition behavior between different phases, after which the mass spectrometer ionizes the separated components and determines their molecular weights and structural features from their motion in electromagnetic fields. The analytical results showed that the clumps contained a specific accelerant component, namely “phosphorus-

based accelerant 300X.” This accelerant is a specialty product of a high-tech chemical company, sold only to a few specific industries, with an extremely limited market circulation. Based on this key clue, the police quickly contacted the company and retrieved the relevant sales records. Investigation revealed that, within the month before the incident, only a small local machinery repair shop had purchased the product. Combined with surveillance footage, witness testimony, and other evidence, the police ultimately identified an employee of that shop as the prime suspect. Investigation confirmed that he had deliberately set the fire because of a personal dispute with the warehouse owner. The successful resolution of the case highlights the important role of analytical chemistry in complex physical evidence identification and the restoration of the truth in criminal cases.

In clinical analysis, analytical chemistry provides key technical support for precise diagnosis and treatment of diseases, especially in the management of chronic diseases. The diagnosis and treatment of diabetes serves as an example. Continuous and accurate blood glucose monitoring is crucial to disease control. A new generation of continuous glucose monitoring systems is based on electrochemical biosensing technology: a miniature sensor implanted under the skin detects the glucose concentration in interstitial fluid in real time. The principle is that glucose undergoes an oxidation-reduction reaction under the action of an immobilized enzyme, producing an electrical signal proportional to the concentration; after acquisition and conversion by the instrument, an accurate blood glucose level is obtained. The system can record a blood glucose reading automatically every 5 minutes, achieving round-the-clock dynamic monitoring and providing both doctors and patients with comprehensive, continuous information on blood glucose changes. Doctors can adjust the hypoglycemic regimen accordingly, and patients can better manage their diet, exercise, and medication,

thereby effectively controlling blood glucose fluctuations, delaying complications, and improving quality of life. Analytical chemistry lays a reliable data foundation in this process, significantly enhancing the refinement and personalization of diabetes management.

In environmental analysis, analytical chemistry provides key technical support for the identification of environmental pollution and ecological remediation. The pollution control and ecological restoration of the plateau lake watershed in central Yunnan provides an example, where analytical chemistry methods played a core role in monitoring and assessment. For the water bodies of polluted rivers in the region (such as the Haihe and Mingtong rivers), researchers used high-performance liquid chromatography coupled with UV-visible spectrophotometry to perform qualitative and quantitative analysis of organic pollutants in the water, such as polycyclic aromatic hydrocarbons and pesticide residues. This method has high separation efficiency and good accuracy, can effectively resolve complex mixtures, and provides a reliable basis for pollution source identification and the formulation of remediation strategies. At the same time, ion chromatography was used to determine precisely the content of heavy metal ions (such as lead, mercury, and cadmium). This method is highly sensitive and selective; even trace heavy metals can be accurately detected, helping to assess comprehensively the degree of pollution of the water bodies and the potential ecological risks. In addition, source apportionment techniques such as factor analysis and multiple linear regression were used to systematically identify the main pollution sources of the main stream and tributaries and their contribution rates, enabling precise diagnosis of water environment problems in different river sections. Modern water quality monitoring techniques, such as the monitoring of plastic mass flux in rivers, have also been applied to track the dynamic changes of pollutants in real

time, assess the effectiveness of remediation measures, and provide data support for subsequent strategy adjustment. Through the combined application of these analytical chemistry methods, the pollution control and ecological restoration of the central Yunnan plateau lake watershed have been advanced in a more scientific and precise manner, laying a solid foundation for restoring river ecological functions and improving the environmental quality of the watershed.

Faced with such broad applications, we cannot help but ask a more fundamental question: is analytical chemistry really a science?

1.6 Is Analytical Chemistry a Science?*

In the preceding sections, both the Chinese and English definitions characterize analytical chemistry as a science, which seems self-evident for a discipline that inquires into natural problems. What is the situation in reality, however? In fact, within chemistry itself, and even in the broader scientific community, there is a fairly widespread view that analytical chemistry is essentially a “technique” or “tool,” rather than a “science” in the strict sense.

The origin of this view is not difficult to understand. Analytical chemistry does indeed have a strong instrumental character: it provides measurement means for other disciplines, it solves practical problems, and it depends on the fundamental principles of physics and chemistry. By comparison, theoretical physics and physical chemistry seem closer to the paradigm image of “science.” When a discipline is seen as “serving other disciplines,” its position in the hierarchy of sciences is easily disparaged. This perception affects not only the development of analytical chemistry itself, but also the attitude of readers toward studying the discipline, so it is necessary to clarify the issue. Is analytical chemistry, then, really a science? To answer this question, we must first briefly review the word “science.”

“Science” derives from the Latin “scientia,” meaning “knowledge” or “learning,” and comes from the Latin verb “scire” (to know). In antiquity, “scientia” referred to knowledge acquired through systematic study and learning. In the late eighteenth century, as methods for investigating nature became increasingly systematic, “science” came to designate specifically the natural sciences. In the nineteenth century, with the professionalization and institutionalization of science, “science” gradually replaced “natural philosophy.” In China, the word “科学” was introduced from Japan: Fukuzawa Yukichi translated the Western “science” as “科学” in the early years of the Meiji Restoration; Kang Youwei first used the term in his *Book Catalog of Japan* in 1896; and Yan Fu also adopted this rendering in his translation of *Evolution and Ethics*. Thereafter, “科学” came into wide use in China to denote systematized, professional knowledge and methods of inquiry.

Interestingly, in today’s society, the word “science” has become a highly popular label, borrowed by numerous industries to enhance their own “prestige” and maximize profits, with “wellness science” and “beauty science” being among the most popular with the general public. But are these so-called “wellness sciences” and “beauty sciences” really science? We thus find ourselves in an awkward position: without a unified and clear consensus on “science,” it is difficult to assert whether a discipline possesses scientificity. After all, these so-called “sciences” often invoke certain theories of the natural sciences to endorse themselves, making it extremely difficult to “expose” them. What kind of knowledge, then, counts as true science? We need some criterion by which to judge.

The philosopher of science Popper (Karl Popper), in his critique of logical positivism, proposed the famous demarcation criterion of “falsifiability.” He held that the difference between science and non-science lies not in whether a theory can be verified, for one can always find evidence supporting any theory, but in whether the theory dares to make predictions that can be refuted by facts. In *Conjectures and Refutations* he wrote: “The criterion of the scientific status of a theory is its falsifiability, or refutability, or testability.”

By this criterion, the statements packaged as “wellness science” or “beauty science” often use vague, untestable formulations, or engage only in post-hoc explanation rather than prediction in advance, and therefore do not possess true scientific character. The quantitative theories advanced by analytical chemistry, such as the Beer-Lambert law, the plate theory of chromatography, and the mathematical models of various titration curves, can all be tested by experiment, and are fully open to falsification. If they were incorrect, experimental data would show it. In this sense, analytical chemistry undoubtedly satisfies Popper’s requirement for science.

The falsifiability criterion, however, is not perfect either. Where, after all, does the authority of the scientific community, beyond ordinary knowledge, come from? This requires tracing back to an older belief, one that can be traced to Kant’s conception.

Kant (Immanuel Kant), in *Metaphysical Foundations of Natural Science*, put forward the notion of “science proper,” holding that a discipline can be called a rational science only if it has achieved mathematization. He even asserted: “chemistry can’t be anything more than a systematic art or experimental doctrine, never a science proper, because the principles of chemistry are merely empirical and can’t

be presented a priori in intuition...” In short, in Kant’s view, chemistry, because it lacks mathematization, cannot be regarded as a science proper.

Not coincidentally, the famous physicist Dirac (Paul Dirac) held a similar judgment.

Dirac wrote in “Quantum Mechanics of Many-Electron Systems”: “The underlying physical laws necessary for the mathematical theory of a large part of physics and the whole of chemistry are thus completely known, and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.”^a In short, in Dirac’s view, chemistry should not even be regarded as an independent discipline, and should in due course be absorbed into physics!

^aDirac, P. A. M. *Proc. R. Soc. London, Ser. A* **1929**, 123, 714.

Even today, some well-known figures in the scientific community publicly claim that “chemistry does not exist.” The controversy over whether chemistry can be recognized as a science seems never to have ceased.

If chemistry is so, then what qualification does analytical chemistry, as one of its branches, have to call itself a science? On careful examination of the practice of analytical chemistry, however, we find a fact that is easily overlooked: the reason analytical chemistry is often mistaken for “merely technique” is that its theoretical form differs from that of physics. Physics explains “why,” while analytical chemistry answers “how to measure.” The theory of the latter is built upon the act of “measurement” itself; it is a theory about measurement, not a theory about the objects measured. These are two different kinds of scientific activity: one is a theory of natural phenomena themselves, the other a theory of how to acquire reliable knowledge about natural phenomena.

When we examine analytical chemistry in the light of both Popper's and Kant's criteria, we reach an interesting conclusion: analytical chemistry stands firm on both scales. By Popper's falsifiability criterion, the quantitative theories of analytical chemistry can be subjected to experimental testing; by Kant's mathematization criterion, analytical chemistry has also been continuously evolving in that direction: the establishment of titration curve equations, the calculation of endpoint errors, and the quantitative description of indicator color-change ranges already constitute a complete mathematical framework.

An important example is the 2020 paper "Hard modeling of titration curve for absolute analysis," which, starting from basic chemical reaction principles and through rigorous mathematical deduction, established a mathematical model for absolute quantitative analysis. This means that under certain conditions, quantitative results can be obtained directly from first principles without relying on calibration by reference standards. This shows that even in the most classical field of chemical quantitative analysis, analytical chemistry is forming its own mathematical theoretical system.

There is more than one form of science. As "the science of measurement," analytical chemistry builds its theory on the logic of the act of measurement, not merely on a pile of empirical rules. It satisfies the requirement of falsifiability, and is also advancing along the road of mathematization. It is in this sense that defining the chemical quantitative analysis part of analytical chemistry as science has its rationale. The instrumentality of a discipline does not preclude its scientificity; quite the contrary, providing a reliable measurement cornerstone for the whole edifice of science is itself a scientific undertaking.

Self-Check

Reflection questions: Kant held that “only a discipline that has achieved mathematization can be called a rational science.” Please answer:

1. Which contents of analytical chemistry have already achieved mathematization? Give an example mentioned in this chapter.
2. In your opinion, which parts of analytical chemistry have not yet been fully mathematized? Does this constitute a fundamental challenge to the “scientificity” of analytical chemistry? Briefly state your view.

Chapter Summary

1. **Definition of analytical chemistry.** Analytical chemistry is a science that develops and applies various methods, instruments, and strategies to obtain information about the composition and properties of matter, within the dimensions of time and space. “Analytical chemistry” differs from “chemical analysis”: the latter is the technical activity of applying existing analytical methods to solve concrete problems, while the former includes the construction of abstract theory and method innovation.
2. **The general process of analytical chemistry.** A complete analytical undertaking comprises six stages: problem definition and method selection, sampling, sample preparation, measurement, data processing and analysis, and reporting and interpretation of results. The process has a feedback-iteration character and is not completed in a single one-way pass.
3. **Technique → method → procedure → protocol.** A technique is an analytical principle, a method is the concrete implementation of a technique, a procedure is a guide to applying a method in a particular setting, and a protocol is a procedure that has been standardized and officially recognized. The four constitute a complete chain from theory to practice.

4. **Classification of analytical methods.** By method principle, they are divided into chemical analysis methods (such as gravimetric and titrimetric methods) and instrumental analysis methods (such as spectroscopic, chromatographic, electrochemical, and mass spectrometric methods); by task, into qualitative analysis, quantitative analysis, structural analysis, speciation analysis, and energy-state analysis.
5. **Criteria for choosing an analytical method.** The choice of an analytical method requires a comprehensive trade-off among accuracy/precision, sensitivity/detection limit, selectivity, speed of analysis, cost, sample size requirements, degree of automation, and other criteria, and these criteria often trade off against one another.
6. **The history of analytical chemistry.** From the empirical observations of ancient fire assaying and alchemy, to Lavoisier's law of conservation of mass, Dalton's atomic theory, and Berzelius's atomic weight determinations, and then to the revolutionary breakthroughs of spectroscopy, chromatography, and mass spectrometry, the evolution of analytical chemistry runs through the course of human civilization. China has made important contributions in Ge Hong, alchemy, and modern mass spectrometry and chromatography.
7. **The broad applications of analytical chemistry.** From pharmaceutical analysis (the thalidomide incident), biomedicine (the Human Genome Project), food safety, industrial manufacturing, and forensic identification to environmental monitoring, analytical chemistry is the core bridge connecting scientific theory with real-world needs.
8. **The scientificity of analytical chemistry.** Kant held that only a discipline that has achieved mathematization can be a rational science, a criterion that once posed a challenge to the disciplinary status of analytical chemistry. However, advances such as the plate theory of chromatography, chemometrics, and the mathematical modeling of absolute quantita-

tive analysis show that analytical chemistry is forming its own mathematical theoretical system, and its scientificity is increasingly confirmed.

Exercises

1.1 Explain in your own words the difference between “analytical chemistry” and “chemical analysis.” Can you find, from daily life, two examples of activities belonging respectively to “analytical chemistry” and “chemical analysis”?

1.2 Describe the general process of analytical chemistry. If you need to determine the content of lead in a pond water sample, explain how you would complete each stage of the process in turn.

★★ 1.3 A food testing agency needs to choose one of the following two methods for determining trace organophosphorus pesticide residues in fruit:

- Method A: gas chromatography-mass spectrometry (GC-MS), detection limit $0.01 \mu\text{g/kg}$, cost about 500 yuan per analysis
- Method B: enzyme inhibition rapid test, detection limit 0.1 mg/kg , cost about 5 yuan per analysis

Using the method selection criteria learned in this chapter, analyze which method should be preferred in each of the following scenarios, and give the reasons:

- (a) A supermarket conducts rapid screening of fruit on each day's shipments
- (b) A provincial food and drug testing institute performs statutory arbitration testing of suspected noncompliant samples

★★ 1.4 Choose a figure from the history of analytical chemistry that interests you (such as Boyle, Lavoisier, Dalton, Berzelius, or Bunsen), consult extracurricular

materials about his life and specific contributions to analytical chemistry, and write a short essay of no fewer than 800 words.

★★ 1.5 Kant's criterion of the "rational science" holds that a discipline can be called science only if it has achieved mathematization. Based on what you have learned in this chapter, do you think analytical chemistry satisfies this criterion? Give your judgment and arguments.

★★ 1.6 In March 2021, President Xi Jinping emphasized at the ninth meeting of the Central Committee for Financial and Economic Affairs of the Chinese Communist Party that "carbon peak and carbon neutrality should be incorporated into the overall layout of ecological civilization." What roles do you think analytical chemistry can play in achieving the "dual carbon" goals? Give at least two concrete scenarios.

1.7 Classify the following items into the levels of "technique (technique) → method (method) → procedure (procedure) → protocol (protocol)," and briefly state the reason for each classification:

- (a) The Work Instruction for Determination of Lead in Surface Water by Graphite Furnace Atomic Absorption Spectrometry compiled by a provincial environmental monitoring station
- (b) The principle of determining lead in water samples by graphite furnace atomic absorption spectrometry (characteristic absorption of 283.3 nm light by lead atoms)
- (c) National Standard GB/T 5750.6-2006 *Standard Examination Methods for Drinking Water, Metal Parameters*
- (d) A concrete analytical scheme for the determination of lead in a reservoir source water (containing humic acid interference) by nitric acid-

perchloric acid digestion followed by graphite furnace atomic absorption spectrometry

1.8 In the melamine incident at the beginning of this chapter, analytical chemists confirmed the presence of melamine in milk powder. Following the six stages of the general process of analytical chemistry, reconstruct the complete workflow of this analytical undertaking. Do you think the process involved feedback and iteration? If so, at which stage might it have occurred?

★★ 1.9 Analytical chemistry is divided into chemical analysis methods and instrumental analysis methods according to method principle. For the following analytical tasks, which class of methods would you prefer? Give the reasons.

- (a) Determining the total hardness (calcium and magnesium) of tap water (estimated content about 50 mg/L)
- (b) Determining the content of arsenic in a traditional Chinese medicinal herb (limit standard 2 mg/kg)
- (c) Monitoring the real-time emission concentration of sulfur dioxide in the flue gas of a steel plant

1.10 The table below lists five scientists in the history of analytical chemistry and their contributions. Match each scientist with the correct contribution, and arrange them in chronological order.

1.11 This chapter divides analytical tasks into five types: qualitative analysis, quantitative analysis, structural analysis, speciation analysis, and energy-state analysis. Classify the following analytical work into the most appropriate type:

- (a) Using infrared spectrometry to identify whether an unknown white powder is heroin

Scientist	Contribution
Berzelius	Proposed the law of conservation of mass and listed a systematic table of 33 elements
Tswett	Determined the atomic weights of more than 40 elements and perfected quantitative analysis procedures
Lavoisier	Founded spectral analysis with Bunsen and discovered cesium and rubidium
Kirchhoff	Separated plant pigments with a chalk column and invented chromatography
Fresenius	Published <i>Introduction to Qualitative Chemical Analysis</i> and founded the first analytical chemistry journal

- (b) Using EDTA titration to determine the content of calcium carbonate in an antacid
- (c) Using HPLC-ICP-MS to determine the respective contents of trivalent and pentavalent arsenic in soil
- (d) Using X-ray single-crystal diffraction to determine the three-dimensional configuration of a newly synthesized anticancer compound
- (e) Using photoelectron spectroscopy to study the change in surface electron-state density of a lithium battery cathode material

★★ 1.12 In the “Minamata disease” incident in Japan, the toxicity of methylmercury was far greater than that of inorganic mercury. Based on the content of this chapter on speciation analysis, answer the following questions:

- (a) If a testing agency reports only “total mercury content of $0.5\text{ }\mu\text{g/L}$ ” for a water body, without reporting the methylmercury content, is the report sufficient for assessing the ecological risk of the water body? Why?
- (b) Give another element for which “speciation analysis” is important, and

briefly explain the reason.

★★ 1.13 A water quality testing laboratory needs to purchase an analytical instrument for routine screening of heavy metals in drinking water (with the requirement of simultaneously detecting lead, cadmium, copper, zinc, and other elements, and a detection limit at the $\mu\text{g/L}$ level). Three options are available:

- (a) Inductively coupled plasma mass spectrometer (ICP-MS), unit price 1.5 million yuan, about 80 yuan per analysis, capable of simultaneously detecting 30+ elements
- (b) Flame atomic absorption spectrometer (FAAS), unit price 200,000 yuan, about 15 yuan per analysis, one element at a time
- (c) UV-visible spectrophotometer (UV-Vis), unit price 50,000 yuan, about 5 yuan per analysis, requiring separate color development for each element

Using the method selection criteria of this chapter (accuracy, sensitivity, selectivity, speed, cost, degree of automation, sample size, etc.), analyze which option you recommend for the need of “routine screening of heavy metals in drinking water,” and state the reasons.

1.14 Briefly explain the meanings of the following terms, and describe the hierarchical relationship among them:

technique, method, procedure, protocol. Using the gravimetric determination of sulfate by barium carbonate as an example, illustrate how each of the four concepts is embodied in this instance.

Answer

1.1 Analytical chemistry is a science (developing and applying methods, instruments, and strategies); chemical analysis is a technical activity (applying existing methods to solve concrete problems).

1.2 Six stages: problem definition and method selection → sampling → sample preparation → measurement → data processing and analysis → reporting and interpretation of results. Feedback and iteration are possible at every stage.

1.3 (a) Method B (enzyme inhibition test), because it is fast and low-cost, suitable for preliminary screening. (b) Method A (GC-MS), because of its high sensitivity and good selectivity, suitable for statutory arbitration.

1.4 Open-ended writing exercise; no standard answer.

1.5 Open-ended discussion question; no standard answer.

1.6 For example: (1) greenhouse gas emission monitoring; (2) quantitative assessment of carbon sinks; (3) carbon footprint analysis; (4) characterization of clean energy materials. Give any two.

1.7 (b) technique → (d) method → (a) procedure → (c) protocol.

1.8 The six stages proceed in order; feedback and iteration may occur between the measurement and data processing stages (such as optimizing sample preparation when recovery is abnormal), or at the reporting stage (such as replanning the strategy when results do not meet expectations).

1.9 (a) Chemical analysis method (EDTA titration), because it is low-cost, simple, and sufficiently accurate. (b) Instrumental analysis method, because of the trace-level detection requirement, for which chemical analysis detection limits are insufficient. (c) Instrumental analysis method, because real-time on-line monitoring is required.

1.10 Lavoisier → conservation of mass; Berzelius → atomic weights; Fresenius → textbook and journal; Kirchhoff → spectral analysis; Tswett → chromatography. In chronological order: Lavoisier (1789) → Berzelius (1807-1818) → Fresenius (1841) → Kirchhoff (1859) → Tswett (1906).

1.11 (a) qualitative analysis; (b) quantitative analysis; (c) speciation analysis; (d) structural analysis; (e) energy-state analysis.

1.12 (a) Not sufficient. Total mercury includes forms of vastly different toxicity; methylmercury is far more toxic than inorganic mercury and is readily bioaccumulated, so total concentration cannot replace speciation analysis. (b) For example, chromium (Cr): Cr(III) is an essential element, while Cr(VI) is a strong carcinogen. Or arsenic (As): inorganic arsenic is highly toxic, while organic arsenic (such as arsenobetaine) is of low toxicity.

1.13 Recommend Option 1 (ICP-MS). Reasons: (1) it can simultaneously detect 30+ elements, giving high throughput; (2) its detection limit reaches the ng/L level, satisfying the requirement; (3) multi-element overall cost-effectiveness is optimal. Option 2 (FAAS) can also meet the detection limit but has low throughput; Option 3 (UV-Vis) has an insufficient detection limit.

1.14 Technique: the precipitation reaction principle → method: the concrete scheme established for a certain water sample → procedure: the operating details compiled by a laboratory → protocol: the national standard.

Index

- accuracy, 32
- alchemy, 36
- Berzelius, 44
- biomedical analysis, 63
- Boyle, 42
- chemical analysis, 17
- chemical analysis methods, 26
- chromatography, 27
- classical analytical chemistry method, 26
- clinical analysis, 63
- Dalton, 43
- detection limit, 32
- electrochemical method, 27
- energy state analysis, 30
- environmental analysis, 63
- food analysis, 63
- forensic analysis, 63
- Fresenius, 48
- Ge Hong, 38
- gravimetric analysis, 49
- HPLC, 17
- human genome project, 64
- industrial analysis, 63
- instrumental analysis methods, 26
- Kjeldahl method, 16
- Lavoisier, 42
- mass spectrometry, 27
- melamine incident, 16
- method, 22
- Minamata disease, 30
- Nernst equation, 51
- pH, 53
- pharmacological analysis, 63
- precision, 32
- procedure, 22
- protocol, 22
- qualitative analysis, 28
- quantitative analysis, 29
- sample preparation, 21
- sampling, 21

selectivity, [32](#)

sensitivity, [32](#)

species analysis, [30](#)

spectroscopic method, [27](#)

structural analysis, [29](#)

technique, [22](#)

thalidomide, [63](#)

thermal analysis, [27](#)

References

- [1] D. A. Skoog; D. M. West; F. J. Holler; S. R. Crouch. Fundamentals of Analytical Chemistry. Cengage Learning, Inc. 2014.
- [2] D. S. Hage; J. D. Carr. Analytical Chemistry and Quantitative Analysis. Pearson Education, Inc.. 2011.
- [3] D. Harvey. Modern Analytical Chemistry. McGraw-Hill Higher Education. 2000.
- [4] 甘峰. 分析化学基础教程. 北京: 化学工业出版社. 2007.
- [5] 武汉大学主编. 分析化学. 北京: 高等教育出版社. 2018.
- [6] 崧诗松; 程依明; 濮晓龙. 概率论与数量统计教程. 高等教育出版社. 2015.
- [7] A. Ringbom. Complexation in analytical chemistry. R. E. Krieger Pub. Co.. 1979.
- [8] A. Ringbom; 戴明. 分析化学中的络合作用. 北京: 高等教育出版社. 1987.
- [9] L. Rayleigh. On an Anomaly encountered in Determinations of the Density of Nitrogen Gas. Proceedings of the Royal Society of London. 1894. 55. 340–344.
- [10] W. S. Gosset. The Probable Error of a Mean. Biometrika. 1908. 6(1). 1–25.
- [11] P. P. von Weimarn. The Precipitation Laws.. Chemical Reviews. 1925. 2(2). 217–242.

- [12] H. H. Willard; N. K. Tang. A Study of the Precipitation of Aluminum Basic Sulfate by Urea. *Journal of American Chemical Society*. 1937. 59(3). 1190–1196.
- [13] G. Gran. Determination of the equivalent point in potentiometric titrations. *Acta Chemica Scandinavica*. 1950. 4. 559–577.
- [14] B. P. Kibble. A Measurement of the Gyromagnetic Ratio of the Proton by the Strong Field Method BT - Atomic Masses and Fundamental Constants vol 5. in (ed Sanders, J H and Wapstra, A H), Springer US, 545–551. 1976.
- [15] R. Kellner. Education of analytical chemists in Europe. The WPAC Eurocurriculum on analytical chemistry. *Analytical Chemistry*. 1994. 66(2). 98A–101A.
- [16] A. T. Woolley; R. A. Mathies. Ultra-High-Speed DNA Sequencing Using Capillary Electrophoresis Chips. *Analytical Chemistry*. 1995. 67. 3676–3680.
- [17] A. T. Woolley; G. F. Sensabaugh; R. A. Mathies. High-Speed DNA Genotyping Using Microfabricated Capillary Array Electrophoresis Chips. *Analytical Chemistry*. 1997. 69. 2181–2186.
- [18] R. de Levie. General expressions for acid-base titrations of arbitrary mixtures. *Analytical Chemistry*. 1996. 68(4). 585–590.
- [19] R. de Levie. A general simulator for acid-base titrations. *Journal of Chemical Education*. 1999. 76(7). 987.
- [20] 甘峰. 基于 EDTA 的配位滴定通式. *大学化学*. 2007. 22(5). 54–58.
- [21] 黎朝. 氧化还原滴定计算新思路. *大学化学*. 2013. 28(6). 66–70.

- [22] 闭凤丽; 尹华勤; 甘峰. 滴定分析终点误差计算公式之探讨. 化学通报. 2015. 78(9). 859–863.
- [23] 李莲云; 余惠娟; 尹华勤; 甘峰. 酸碱滴定中指示剂浓度的影响. 大学化学. 2015. 30(6). 79–82.
- [24] J. Tellinghuisen. Calibration: Detection, Quantification, and Confidence Limits Are (Almost) Exact When the Data Variance Function Is Known. Analytical Chemistry. 2019. 91(14). 8715–8722.
- [25] 夏锦尧. 实用荧光分析法. 北京: 中国人民公安大学出版社. 1992.
- [26] M. Anastassiades; S. J. Lehotay; D. Štajnbaher; F. J. Schenck. Fast and Easy Multiresidue Method Employing Acetonitrile Extraction/Partitioning and “Dispersive Solid-Phase Extraction” for the Determination of Pesticide Residues in Produce. Journal of AOAC International. 2003. 86(2). 412–424.